

EARLY FIRE DETECTION IN FOREST USING IMAGE PROCESSING



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Submitted to the Department of Computer Engineering in the
partial fulfillment of the requirements for the degree of
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**Department of Computer Engineering
Bahria University Islamabad
2025**

CERTIFICATE

EARLY FIRE DETECTION IN FOREST USING IMAGE PROCESSING

SDG No	Description of SDG	SDG No	Description of SDG
SDG 1	No Poverty	SDG 9	Industry, Innovation, and Infrastructure ✓
SDG 2	Zero Hunger	SDG 10	Reduced Inequalities
SDG 3	Good Health and Well Being ✓	SDG 11	Sustainable Cities and Communities ✓
SDG 4	Quality Education	SDG 12	Responsible Consumption and Production
SDG 5	Gender Equality	SDG 13	Climate Change ✓
SDG 6	Clean Water and Sanitation	SDG 14	Life Below Water
SDG 7	Affordable and Clean Energy	SDG 15	Life on Land ✓
SDG 8	Decent Work and Economic Growth	SDG 16	Peace, Justice and Strong Institutions
		SDG 17	Partnerships for the Goals



Range of Complex Problem Solving			
	Attribute	Complex Problem	
1	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues.	✓
2	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.	✓
3	Depth of knowledge required	Requires research-based knowledge much of which is at, or informed by, the forefront of the professional discipline and which allows a fundamentals-based, first principles analytical approach.	✓
4	Familiarity of issues	Involve infrequently encountered issues	
5	Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering.	
6	Extent of stakeholder involvement and level of conflicting requirements	Involve diverse groups of stakeholders with widely varying needs.	
7	Consequences	Have significant consequences in a range of contexts.	✓
8	Interdependence	Are high level problems including many component parts or sub-problems	✓
Range of Complex Problem Activities			
	Attribute	Complex Activities	
1	Range of resources	Involve the use of diverse resources (and for this purpose, resources include people, money, equipment, materials, information and technologies).	✓
2	Level of interaction	Require resolution of significant problems arising from interactions between wide ranging and conflicting technical, engineering or other issues.	
3	Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways.	✓
4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.	✓
5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches.	

UNDERTAKING

I certify that research work titled “*Early Fire Detection in Forest Using Image Processing*” is my own work. The work has not been presented elsewhere for assessment. Where material has been used from other sources, it has been properly acknowledged/referred.

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01-132212-031

DEDICATION

This research is dedicated to the survivors of forest fires who have endured immense human and environmental losses. Their resilience and persistence in the face of such devastating circumstances have been a constant inspiration for this project. We hope that our work contributes to the development of tools that can prevent future wildfires and help protect both lives and nature.

Our goal is to create a reliable, scalable, and cost-effective early fire detection system that can assist in reducing the impact of forest fires on communities and ecosystems. Through innovation and collaboration, we aim to save lives, preserve the environment, and contribute to the global efforts in wildfire prevention and management.

We express our heartfelt gratitude to our families, particularly our parents, whose unwavering support, encouragement, and belief in us have been instrumental throughout this journey. Their love and guidance have been our foundation.

We also extend our thanks to our friends and siblings, whose companionship and understanding have been a source of strength throughout this journey.

ACKNOWLEDGEMENTS

To begin with, we offer our heartfelt thanks to Allah Almighty, the Creator of all life, the source of our intellect, and the giver of strength. It is through His endless blessings and guidance that we have been able to complete this project. We are deeply grateful for the opportunities and resources provided to make this research possible.

We extend our sincere appreciation to **Engr. Muhammad Kashif Naseer**, whose unwavering support and expert guidance were invaluable throughout the entire course of this project. His vast knowledge, insightful suggestions, and constant encouragement helped us overcome every challenge, and his contributions were pivotal to the success of this research.

Our deepest gratitude also goes to our families, whose belief in us and constant support were a driving force behind our determination. We are especially thankful to our parents, whose unwavering faith in our abilities and sacrifices gave us the strength to persevere and achieve our goals.

We also thank all personnel and institutions, who contributed their efforts in the field of fire detection systems, which contributed to the realization of our project.

ABSTRACT

Early fire detection stays vital because forest fires threaten natural ecosystems as well as local wildlife population and nearby residential areas. Fire detection methods that depend on manual patrolling and satellite imaging along with thermal sensors tend to be expensive yet offer slow responses with unpredictable precision levels. The proposed system implements a real-time forest fire detection mechanism through image processing with deep learning technology that combines Raspberry Pi 4B and Pi Camera and aerial drone surveillance. The You Only Look Once (YOLO)v5s object detection model within the system performs accurate early fire signal detection from drone image acquisition. The detection triggers the system to determine exact position information with a Global Positioning System (GPS) module that sends location data to a Firebase web platform. The solution enables automatic alerts to emergency teams. The system demonstrates cost-efficient operation along with scalability features and quick response times which allows it to work continuously in forest areas and high-risk zones. Embedded systems, aerial surveillance, Artificial Intelligence (AI) technology together with real-time reporting within the recommended system design boost forest fire detection as well as emergency response speed.

Keywords: Early Fire Detection, Image Processing, YOLOv5s, Raspberry Pi, Pi Camera, Drone Surveillance, GPS, Real-Time Monitoring, Firebase Portal.

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CHAPTER 1

Introduction

Forests are the most vital ecosystem on earth because they provide social, economic, and environmental benefits. These habitats play a crucial role in preventing climate change through their CO₂ absorption and oxygen release functions, while influencing atmospheric weather patterns. Multiple plant and animal species find their habitats in forests, which creates conditions for biodiversity preservation and ecological stability. The economic worth generated from forests reaches substantial levels when forest products such as timber join forces with natural medicines and recreational tourism for financial gain. The diverse domains including climate regulation, along with biodiversity conservation, sustainable resource management and ecosystem services form valuable ecosystems that now faces an increasing severity of forest fires [1], [2]. These interconnected functions demonstrate the essential role forests play in sustaining life while maintaining environmental stability thus requiring their protection for human survival and ecological stability.

Natural forest fires have existed for numerous years, but human-caused environmental changes have led to a rise in their severity. The annual forest burnings reported by Global Forest Watch amount to 4.2 million hectares of forest area [2]. Forest fires endanger wildlife while hurting local communities and generate vast carbon dioxide air emissions through their destructive processes. The states of California, Amazon rainforest and Australia suffer from extensive wildfires which create billions of dollars of financial damage.

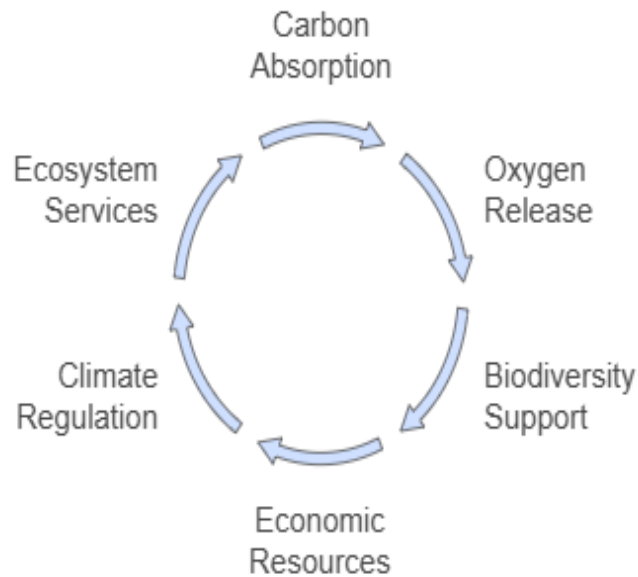


Fig 1.1: The Forest Ecosystem Cycle

The Fig 1.1 demonstrates an endless repetition of fundamental ecological operations. Through the process of carbon absorption and oxygen release, plants maintain biodiversity. The numerous forms of life in the region provide instrumental economic resources while assisting in climate regulation. Human activities that keep ecosystems in good condition provide essential ecosystem services, which enhance carbon uptake mechanisms that maintain environmental stability [2].

1.1 Forest Fire Impact and Challenges

Forest fires are hazardous events that significantly impact species, ecosystems, and the climate. They result in the loss of critical ecosystems, where thousands of species are displaced and biodiversity is harmed. Additionally, they emit high amounts of carbon dioxide and other greenhouse gases into the atmosphere, which fuels global warming and accelerates climate change [1], [2]. The ripple effects of wildfires on forests not only affect the environment but also the livelihoods of people who depend on forests for their food and income.

1.1.1 Forest Fires in Pakistan

For Pakistan, wildfire is emerging as a major threat especially to areas that feature distinctive ecological systems and exceptional biodiversity. The areas under attack include:

- **Balochistan:**

Sherani Forests renowned for its Juniper trees, this region faced devastating wildfires in 2022, which ravaged 1,500 hectares of forest [3]. Ziarat District having Juniper woodlands with a history spanning hundreds of years, are often at risk from fires [4], [5]. Hingol National Park, this area is incredibly susceptible to fire due to its arid climate and sparse vegetation.

- **Khyber Pakhtunkhwa (KPK):**

Due to increase in temperature several hundred hectares of Chir Pine forests are annually burnt to ashes in the Swat Valley [6]. Abbottabad and Mansehra characterized by big Pine and Deodar forests, it is not a rare occurrence in these regions when wildfires are reported during periods of drought or because of human negligence [4], [6]. The effects of climate change are increasingly making Alpine and subalpine forests vulnerable in Chitral and Dir Districts.

- **Margalla Hills (Islamabad):**

Human activities cause a fire, near an urban area, resulting in a major decrease in the biodiversity index and rising air pollution levels in the Islamabad. Wildfires are becoming prone due to climatic change, human carelessness, and dry climatic conditions [7], [8]. In various places, it leads to permanent ecological damage, decreased water holding capacity, and soil erosion. Besides, carbon is emitted as well, which results in global warming and disorganizes local weather systems, contributing to the toll on the environment. Thus, there is a dire need for targeted and innovative early detection system, as well as effective forest fires management to deal with the above challenges [9], [10].

1.1.2 Background of Forest Fire Management

Forest fires have occurred naturally for centuries; however, their intensity has significantly increased due to changes in the environment and human activity.

According to Global Forest Watch, around 4.2 million hectares of forest are burned annually [2]. These fires not only pose a threat to the wildlife and local people but also cause immense damage as they emit large quantities of carbon dioxide into the atmosphere [11]. Regions such as California, the Amazon rainforest, and Australia experience massive wildfires that result in damages costing billions of dollars [9], [12], [13].

Areas in Pakistan that are particularly prone to forest fires include Khyber Pakhtunkhwa, especially at locations such as Shogran, Swat, and the Malakand Division, Balochistan where the Ziarat Juniper Forest and Koh-i-Suleiman are significantly exposed and Margalla Hills National Park [4], [14], [15], [5]. These areas lack proper fire detection systems and management services.

For instance, in the Sherani District of Balochistan in 2022, a disaster devastated over 1,500 hectares of land, the environment and inflicted loss for the economy as well [3]. In June 2022, Margalla Hills faced one major blaze fire, escalated with the lack of quick response to control this fire [7]. All such emergencies need the introduction of novel, reliable, and relatively economical technologies that detect such fires early in time.

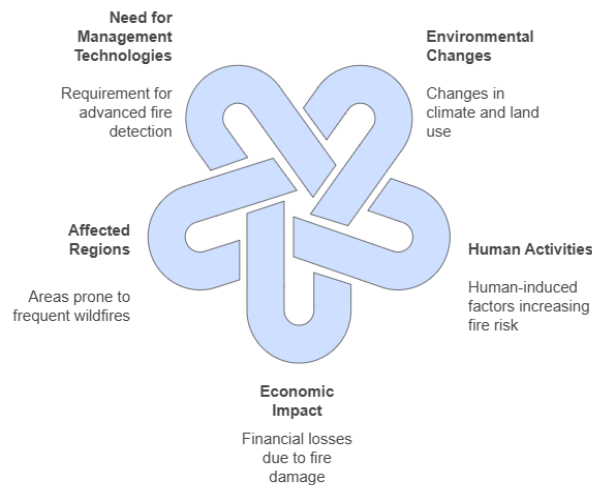


Fig 1.2: Components of forest fire challenges

The Fig 1.2 illustrates the key components contributing to forest fire challenges, which include environmental changes and human activities resulting in economic loss from fire damage, and regions prone to frequent wildfires [2], [3].

1.1.3 Annual Area Burnt by Wildfires in Pakistan

The Fig 1.3 [2] shown below indicates that countries without resources, such as Pakistan, where firefighting infrastructure is often insufficient, urgently need cost-effective and reliable fire detection and control systems.

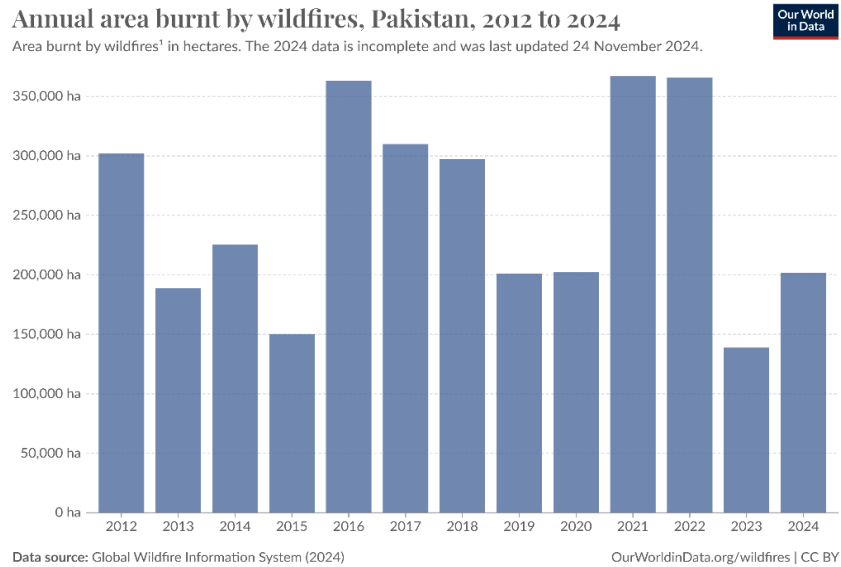


Fig 1.3: Global Wildfire Information System (2024), Our World in Data [2]

1.2 Existing Solutions and Their Limitations

1.2.1 Current Systems

Visible Infrared Imaging Radiometer Suite (VIIRS) and Moderate Resolution Imaging Spectrometer (MODIS) have the potential to cover vast areas. Nevertheless, they are characterized by some drawbacks such as low-resolution images, these technologies respond slowly and react late [10], [16]. These drones are costly and have a low range, but they can sense fires accurately because of thermal imaging cameras [17]. It is expensive to establish ground-based sensors in remote locations as they require an extremely high-density network for temperature and humidity monitoring [18]. The

precision, cost, and effectiveness of different worldwide technologies used to detect the outbreak of forest fires significantly vary. Some of these are discussed in Fig 1.4 given below:

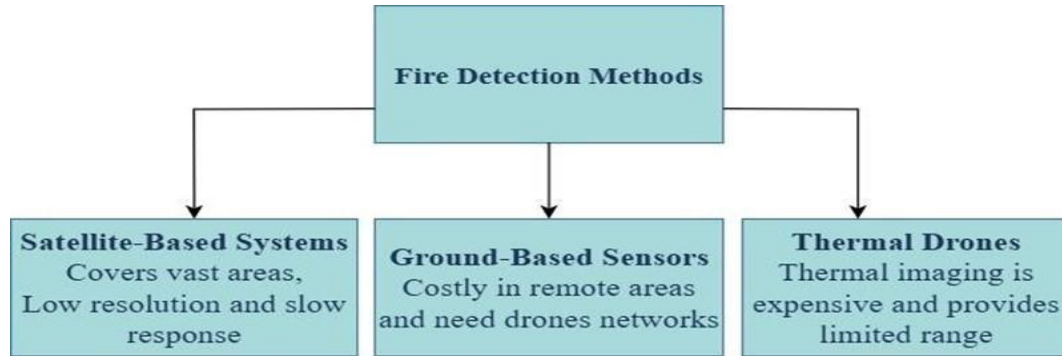


Fig 1.4: Present Technologies for Forest Fire Detection

1.2.2 Limitations

While each system has its own benefits, they all share several common disadvantages:

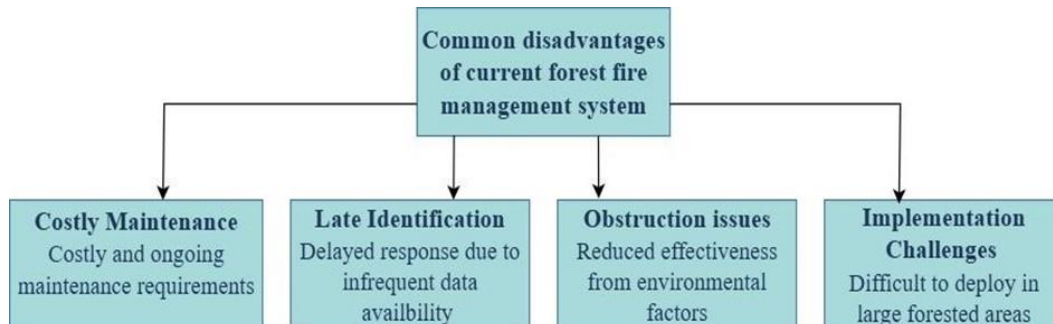


Fig 1.5: Disadvantages of Current Forest Fire Management Systems

In Fig 1.5, current forest fire detection systems are expensive, require frequent maintenance and have delayed detection. Their implementation is difficult in vast forested areas, and they are not ideal for remote regions due infrastructure dependency.

1.3 Problem Statement

Forest fire causes a serious threat to biodiversity, natural resources, and human safety. While early fire detection is important, existing methods such as satellite imaging, thermal drones, ground sensors, and human surveillance have notable limitations, including high costs, delayed detection, limited coverage and environmental barriers

like smoke and rough terrain. These challenges cause timely and effective intervention, especially in remote and vulnerable forest regions. There is an urgent need for a cost-effective, scalable and accurate system capable of real-time fire detection with minimal infrastructure dependence.

1.4 Proposed Solution

The proposed framework utilizes Raspberry Pi Camera Module with deep learning methods to achieve a low-cost detection system of early fire signs instead of expensive thermal cameras or satellite imagery systems. Raspberry Pi camera enables this system to operate at low cost in areas where resources are limited, such as Pakistan.

1.4.1 Key Features

The main key features of the proposed project are:

- **Drone Integration:** Drone integration provides mobility to monitor unreachable places.
- **GPS Tracking:** GPS helps to track locations of fires.
- **Deep Learning (DL) Techniques:** Ensure accuracy at the early stages of detecting a early fire signs.
- **Fire and smoke:** The system is capable of identifying both fire and smoke. Smoke detection makes the system more efficient because it is often visible before fire, allowing earlier warnings.
- **Real-time Monitoring:** Drone equipped with camera take images of live activities that are processed in real-time immediately by deep learning models. This makes the fire and smoke detection instantaneous lowering the time of responding and removes the aspect of manual monitoring.

1.5 Objectives

The core objective of this work as shown in Fig 1.6 involves building an early fire detection system for forests using image processing technology, which must be cost-effective and efficient at scale. The system uses a drone equipped with a Raspberry Pi camera to capture images, then applies deep learning models on these images to detect

fire and smoke in early stages. A web portal operates as a user-friendly interface that allows authorities to view and control alerts instantly to decrease response time. The project avoids satellite systems, thermal drones and ground sensors to achieve cost-effective and real-time solution that works best in different forest environments, particularly in developing countries such as Pakistan.

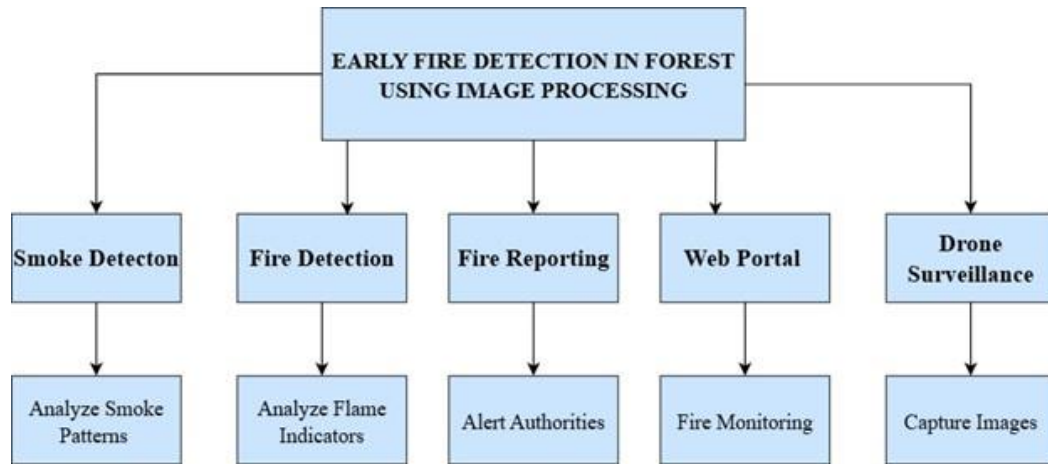


Fig 1.6: Project Goals

CHAPTER 2

Literature Review

This chapter contains literature review associated with our proposed project “Early Fire Detection in Forest using Image Processing”. The problem is identified as the realization of the current approaches and technologies that aid the early detection and broadcasting of the threat of fire in forests. Forest fires remain the high threat to the ecosystem, wildlife, and neighbors living well around, so it is the necessity to find out efficient ways to timely detect and respond to forest fires. Historical methods of fire detection usually present problems, including delayed notifications, poor coverage, high expenses and failure to show actual time monitoring. The modern monitoring systems have been feature changing in the recent past to come up with more responsive and efficient ways of overcoming such limitations. These systems are based on automated functionality of monitoring the environmental changes and other symptoms of an emerging fire like the occurrence of smoke or increase in heat levels. Furthermore, communication systems have been incorporated so that any alerts and essential information is transmitted immediately to the authorities that are concerned. This also assists in enhancing decision making, coordination and response in emergency. The literature review can be used to emphasise the changes to the systems, their effectiveness, and the gaps and the opportunity to improve them that can be utilised during the implementation of our proposed solution.

2.1 Related Studies

2.1.1 Drone Network for Early Warning of Forest Fire and Dynamic Fire Quenching Plan Generation

The paper begins with details of a next-generation system for forest fire detection and management using Unmanned Aerial Vehicle (UAVs), IoT and deep learning. Real-time fire identification and tracking are made possible by UAVs with thermal and Red Green Blue (RGB) cameras along with Internet of Things (IoT) sensors that measure

environmental parameters to identify potential fire threats [19]. The system employs GoogLeNet-TL for fire detection with 96% accuracy and F1-score of 97% while Particle Swarm Optimization (PSO) is employed for drone path planning [22],[23]. Extension of knowledge enhances the efficiency of the given models, and IoT integration provides timely alert for fire prevention and its handling.

Depending on the type of sensors employed in the system, mounted on the cameras, and particularly on thermal ones, the price of the system raises with potential costs proving to be high during large-scale wildfire monitoring. UAVs are power limited and have a limited flight time and therefore are not suitable for extended applications over large forest areas. This could have consequences in terms of surveillance during long manned missions. The system thus relies strongly on communication networks to achieve appropriate integration, which poses a challenge where there is limited connectivity in each region. This can hinder real time exchanges of data and analyses of hatchery programs and the efficiency of the system [24]. Wireless Sensor Network (WSN) via IoT sensors are not accurate in areas with intense environmental conditions, like heat or rainstorm, or windstorms. Further, these systems are not maintenance free and demand frequent service, which means high cost and utilization of resources [25].

Although, the deep learning models have great potential, their usability in real life scenarios comes with a higher demand for computational power for training and for live prediction. It may not be possible in resource limited scenarios or on low power platforms like UAVs [4], [19] or ground operating sensors. False positives alarms are typical for deep learning models, especially when the signals are affected by visually similar conditions, such as fog, smoke, or other fires unrelated manifestations. Optimization tasks within wildfire detection systems provide a need for usage of PSO algorithms where parameters need accurate tuning for getting the best performance [22]. This causes imprecise solutions that will in turn slow down the system depending on the level of tuning done. An issue that arises from this aspect is the ability to scale because there are few diversified datasets that characterize different fire scenes, and environmental factors. Furthermore, high deployment and high training data requirements are challenges that hinder their applicability to a large scale.

2.1.2 Towards Early Forest Fire Detection and Prevention using AI-powered Drones and IoT

The paper describes a preliminary design of a single system which utilizes IoT, AI and drones to support the early detection and prevention of wildfires. Long Range Wide Area Network (LoRaWAN) equipped IoT nodes allow for bulk, low power, monitoring, and drones with AI achieved 99.46% fire detection rate and tracking with particle filters [26]. Drones can be controlled and can track fields and contain putting out mechanisms where cloud support is offered for quicker reactions. Solutions fall under the connectivity architectures of static gateways, mobile drone-borne gateways, and lowly connected sensors with the capability to detect within 1-5 minutes, track effectively, and can easily scale [27]. It is also important to address that IoT devices [28] and cameras are not reliable in extreme conditions such as high temperature, heavy rainfall or windy conditions. This can obviously affect the performance and useful life of the system. Whereas the use of AI models with an emphasis on visually based data, such as roads, weather conditions like smoke and fog or poor visibility may impact the models. These conditions may cause the system to misinterpret or miss fire-related event resulting in lower detection rate.

Some of the disadvantages of UAVs [19] include short battery power that hampers the system's effectiveness when used to conduct survey operations over large regions. Furthermore, they may have direction difficulties just in case they are deployed during strong wind or in haze hence they are not effective in the discovery of wildfires. This system will need an efficient means of passing data back and forth in real time. Where there are no fixed line links available or it is a forest region, the establishment of proper connectivity may sometimes be a problem, which in turn may hamper the smooth flow of information and slow down the system to come into action during fire occurrences as may be expected. The first cost which any agency or organization has to pay to deploy such systems, including the use of drones, AI models, and appropriate sensors, could be very high. This restricts versatility and thus the possibility to expand in large tracts of extensively forested areas.

2.1.3 Forest Fire Detection System using Wireless Sensor Networks and Machine Learning

The paper puts forward a forest fire detection system using WSNs [29], [25] and Machine Learning (ML) for the early identification and fast response. Critical physical characteristics include temperature, humidity, light intensity, and carbon monoxide with solar rechargeable batteries in WSNs providing lifelong power supply in distant locations. The sensor nodes have been developed in an initial manner to endure difficult climatic conditions and animals attacks. A tree topology network is used to transmit data whereby the cluster heads use low energy to forward information to a base station with minimal delay. There is a ML regression model to analyse through collected data to estimate fire occurrences, and the training was conducted under the fire and non-fire circumstances. It schedules alarms for correction by authorized individuals while simplification of unnecessary parts conserves energy for the system [30]. Low light capabilities can cause problems in the functioning of the nodes powered by solar panels. If the training data values are inaccurate, the fire can be either detected when there is none or vice versa. Large forest network architecture based on tree topology is problematic in scalability and robustness. Sensors may still be affected by the elements or animal movement even if measures against adhesive resistance have been implemented. WSNs [25], [30] that are connected to machine learning models may also require deployment and maintenance costs, these are often high. A lack of computational capacity on such devices will also potentially compromise the real-time detection of anomalies.

2.1.4 Fire Detection Using Image Processing

The paper primarily focuses on the fire detection through image processing and applying deep learning models among which is YOLOv3. Compared to the conventional ways of smoke or heat sensors, YOLOv3 uses Convolutional Neural Network (CNN) for fire detection with speed and effectively as it scans the image whole at a time [31]. A new dataset with 1400 images with labels was generated for training the model as there is no complete fire data set available in famous data set like COCO etc [32]. The system averaged 81.76% precision and showed a clear increase

after introducing smoke data. Real time implementation was done using Raspberry Pi in combination with the camera whereby fire detection sent notifications through the Pushover API in order to encourage timely response [33]. Test outcomes including Intersection Over Union (IOU) and Precision Recall showcased the general dependability of the model and a significant increase in performance when tested using more smoke data. It seems this system can be integrated with existing surveillance system for large scale fire detection system. The custom datasets not necessarily reflect real-life conditions especially during extreme weather conditions or when fire intensities are considered. Screen tests revealed low detection rate when the fire was blended with smoke and this was corrected when more smoke information was added. Leveraging a Raspberry Pi for processing reduces computational horsepower which can be a downside to real time results in very complex applications. Differences in lighting or having cluttered backgrounds could provoke completely different results, either false positive or false negative. Large-area networks as part of the system integration's need extra scans involving certain resource and training for proper operation.

2.1.5 Image Fire Detection Algorithms Based on Convolutional Neural Networks

In this paper, a general overview of the CNNs in fire detection from images is presented with the comparison of CNN with that of the conventional fire detecting instruments such as smoke or heat detector [34]. Marine systems in particular are most prone to problems of low accuracy due to size and complexity of the environment in large space. On the other hand, CNNs can extract features of an image automatically and can also detect the features that are intricate and are appropriate for fire detection. This paper discusses the development of some enhanced object detection CNN models such as Faster Region-based Convolutional Neural Network (RCNN), Region-based Fully Convolutional Neural Network (R-FCN), Single Shot MultiBox Detector (SSD) that region proposal to areas that have flames or smoke, improving the networks accuracy and also minimizing on delay when detecting flaming objects. The results showed that YOLOv3 provided the highest accuracy at the detection with an average of 83.7% accuracy and respectable frame rate of 28 Frames Per Second (FPS) which makes it useful for real-time detection [35]. The models were trained on a constructed fire image

dataset and show promise in improving current fire detecting systems, especially in large or in forest and building type areas. Fire conditions can be very versatile, therefore several datasets with high quality are required to describe fire conditions variety, however, traditional sets might lack necessary data.

Similar to other CNN performances, it has the ability to identify false positives and negatives such as those shown in the above YOLOv3 performance due to visible characteristics such as lighting conditions, smoke density and cluttered backgrounds. One of the factors that can subtract the extension of CNNs applications is the demand on computational resources that may be problematic in real-time applying for devices with a small number of computational performance. However, even such enhancements as, for instance, the region proposal function, CNNs may fail in the detection of extremely tiny fires or fires behind the smoke. It still is quite tricky to generalize models across different environments and fire scenarios and more work needs to be done to adapt these models for actual implementations.

2.1.6 Embracing IoT and AI for Forest Protection: Lessons Learned and the Path Forward for the Forest Fire Detection Early Warning System

This article focuses on a relatively recently implemented and ongoing interdisciplinary project that is attempting to design low-cost and easily scalable forest fire detection early warning utilizing AI, Computer Vision, IoT, and Remote Sensing. This smart fire fighting system has been designed through LUMS' collaboration with WWF-Pakistan with the help of KPK Forest Department and uses IoT integrated machines with artificial intelligence, and remote sensing data to detect fires and monitor environmental conditions. The system employs Pan-Tilt-Zoom (PTZ) cameras for video surveillance, which cost less than thermal cameras, alongside IoT sensors, weather stations and smoke detection and LoRa-based communication networks for real-time hazard detection. Moreover, based on the meteorological conditions and remote sensing data machine learning models estimate the fire spreading. The team performed pilot testing on the system controlled environment before implementing in the field areas, while testing the models the system's power components were also designed in order to operate more efficiently and effectively in harsh environments.

The system ranks as being highly flexible and easy to use and provides users with a dashboard that displays current status. Anticipated developments of the system include making it available to the general public through other community established early warning systems and increasing the accuracy of the predictions [36]. Processing of information from numerous sources contributed to the raising of the system's non-triviality of the system and thus introduced decreased efficiency of works. Low cost PTZ cameras and IoT sensors were adopted to create a low-cost scalable solution, however, the system needed to be tuned for electric power consumption, data transmission and making it robust for adverse weather conditions. Changes were made to support decision making and AI at the edge to support data transfer while filtering the necessary information based on power consumption and network traffic. Climate control measures were considered to maintain the system during hot weather, however, there is need to advance in order to adapt the system at different climates. Lack of adequate data to work at district level precipitated issues of inaccurate fire prediction and, therefore, need for extensive data collection. Considering that multiple technologies were to be integrated for system functionality, it was vital to conduct trial and error to make it efficient. However, the team is working on the improvement of the system to expand it and work on the definitions for the fire prediction.

2.1.7 Comparing Machine Learning Algorithms to Predict Vegetation Fire Detections in Pakistan

This study leveraged machine learning classifiers to classify vegetation fires in Pakistan using MODIS Global Fire Atlas data in combination with climatological, topographical and vegetation data from 2001 to 2022. The performance of four machine learning algorithms, namely Logistic Regression, Random Forest, Support Vector Machine (SVM), and Extreme Gradient (XG) boost, was compared to check the accuracy of these models in determining forest, cropland and other vegetation fires. According to the performed comparison of seven machine learning models including Random Forest, the latter yielded the highest level of accuracy that ranges from 78.7% to 87.5% and Area Under the Curve (AUC) values of up to 93.4% for the forest fires. The study also developed fire risk maps, which open up avenues of predicting regions with increased risks of fire incidents in Pakistan [37]. These systems are designed to be used for fire

risk assessment and subsequent prevention by predicting regions likely to catch fire in the future. The available study didn't specify a number of possible issues with the dataset like the absence of data on the provinces where fires occur or the distribution of fires by the time of year. The Random Forest model may be overwhelmed by extraneous liberating factors or by very large fire events that might not have occurred in the past. The model was initially built for forecasting fire risks based on historic occurrence and may not capture the current changes that are liable to alter the climatic and land-use characteristics that propel fires. The applicability of the model may be reduced where it is applied to other areas in the world or other weather conditions. However, the study not only evaluated predictive accuracy but also incorporated and used the models in real functional systems in real time, which were not actually examined in the work, and which can become a possible limitation to implementation.

2.1.8 An Early Forest Fire Detection System Based on DJI M300 Drone and H20T Camera

This paper provides a new approach to solving the problem of early forest fire detection using a Da-Jiang Innovations (DJI) M300 drone with a ZenMuse H20T camera. Thermal and RGB cameras together with Residual Network (ResNet-18) and U-shaped Neural Network (U-Net) make the process of fire detection and confirmation real-time. The drone uses a zigzag pattern with respect to flight and ResNet-18 to classify the images to categories such as 'Normal', 'Potential Fire' or 'Potential Smoke'. Thermal imaging and U-Net segmentation are then used to validate fire and smoke region proposals from the detection step. Geo coordinates are included in real-time alerts provided to ground teams for a quick reaction. During the outdoor experiments, the system was proved to identify, recognize and locate fires, which advocates its usage in operation of wildfires. It also, couples well with Robot Operating System (ROS) and DJI's Software Development Kits (SDKs) to allow the incorporation of hardware and software without problems. The optimized deep learning application models (ResNet-18 and U-Net) yielded final accuracies of 82% and 75% respectively in the fire detection task, indicating that further enhancement of detection precision could still be attained, particularly given more complex fire conditions [38]. The system may experience difficulty in identifying small or emerging fire due to physical conditions

such as; weather conditions, intensity of smoke, and limited vision in producing thermal and visual images. The DJI M300 drone is somewhat restricted due to the type of flights and load its system can provide which might hamper productivity when flying for more extensive forest areas or during conditions such as heavy winds. The on-board computation provided by the NVIDIA Jetson Xavier NX may offer constrained levels of performance especially in terms of speed and power consumption in the larger and more complex missions. Subsequent field tests are, therefore, required to confirm the effectiveness of the system to a similar degree outside of controlled experiments. The use of drones adds another layer of issue to working smoothly, including charging and service requirements that limit constant operations.

2.1.9 Deep Learning and Transformer Approaches for UAV-Based Wildfire Detection and Segmentation

In this work an end-to-end system of wildfire detection and segmentation using UAV is proposed based on deep learning and transformer models. Classification is performed using EfficientNet-B5 and DenseNet-201 with an accuracy of 85.12% and F1-score of 84.77% [39]. For segmentation, three models are employed: TransUNet, TransFire, and EfficientSeg. Another model called TransUNet CNN with transformers garners a 99.9% F1-scores of fire pixels and shapes' identification even amidst a complicated background [39]. The study conducts classification using 48,010 RGB images in the FLAME dataset and segmentation using 2,003 images in the same dataset. The system greatly enhances the area of wildfire identification, which are often small in size and have a challenging background for regular sensors. The integration of the current enhanced AI models with UAVs is a portable, dynamic, and, therefore, financially savvy method for tending to the problem of wildfire [39]. Fire Luminosity and Appearance Metadata for Evaluation (FLAME) dataset which is in RGB color format can pose a challenge to distinguish fires at certain conditions such as during the night or dense smoke. While the models delivered an F1-score of segmentation, their application in the real world may be problematic due to unpredictable conditions and fire sizes. The use of UAVs brings practical concerns such as power constraints on battery-operated UAVs, flight duration and repeated maintenance that may disrupt the constant surveillance of large regions. External conditions such as meteorological

conditions, wind, and conditions at the atmosphere may affect the UAV's manoeuvrability and its accuracy of its sensors in real time thus affecting the system's real time performance. Although the proposed deep learning models achieve high accuracy in simulation, it is still clear that there are improvements needed and all models need to be fine-tuned to perform outstandingly in the various environments and fire situations.

2.1.10 Enhancing Deep Learning-Based Forest Fire Detection with a Diverse Evolving Open-Source Dataset Focused on Data Representativeness and a Novel Multi-Task Learning Approach

In this work, the authors propose a large and broad fire data set of 2,700 RGB images for improved forest fire detection using deep learning techniques. One of the most significant advantages of the dataset is that it pays a lot of attention to representativeness, and covers all types and characteristics of environment, and the presence of interference factors such as clouds and sunlight, a wide range of geographical locations. which is divided into fire and no-fire, with subsidiary classes for smoke and fire and smoke confounder subtypes. The data set is collected from objects from the public images from the fields of government agencies and open site Flickr so that it gives great variability of the terrain, types of forest, weather condition and many more angles of cameras. The paper presents a new Multi-Task Learning (MTL) approach that jointly identifies fire/no-fire instances and eliminates confounders, enhancing the classification model's fire recognition against other visually similar events, such as sunsets and fog. This MTL approach greatly decreases the number of false positives and is more accurate, precise, recall and F1-score efficient than other typical binary classifiers. The two models' performance is compared based on the newly introduced performance indicators the MTL model has a 90.7% accuracy and an F1-score of 88.3%, and the binary classification model's scores are 84% for accuracy and 80.5% for F1 score [40]. This approach offers a high level of feasibility that could be used at large scale in actual wildfire detection systems. Nonetheless, the dataset is quite diverse and contains only 2,700 images, which can hardly be considered enough to let deep learning models learn how the fire scenario may look like and how it is better to react to it in more complex cases. This is especially the case with the type

of area that may not be well-represented in the first place, including for instance, certain terrains and climate conditions.

The use of previously obtained images leave out cases of more localized or variable wildfire situations, as these are gathered from publicly available websites such as Flickr belonging to government organizations. This means that the application of the model is quite limited due to the availability of only RGB images. The performance could be greatly improved with other sensors such as thermal imaging camera during nighttime or in conditions such as thick smoke. When compared to the baseline Incomplete Multimodality Transfer Learning (IMTL), the MTL approach improves the model's discrimination but at the same time also increases computational cost in terms of the amount of resources needed to train and use the model. This could result in real-world dramatic scalability in situations when one has access to little more than basic computing and infrastructure. The study covers false positives and false negatives that could happen in fire detection, the critical issue in wildfire management, which are not fully covered.

2.1.11 Machine Learning Based Early Fire Detection System using a Low-Cost Drone

In this paper, we develop a low power, and low-cost drone system for early forest fire detection using the machine learning and image processing approach. The operating system is based on an X-type UAV with a carbon fiber frame for minimal vibrations and high reliability. Real-time smoke detection, which is often the first indication of fire, is regularly executed by the Raspberry Pi microcontroller with TensorFlow and OpenCV. The ability of the smoke detection model relies on SSD Lite MobileNet that was trained with 280 smoke images dataset. To minimize the false alarms the system uses a number of mechanisms such as image recognition and the post-processing applied to the data obtained through colour modelling and edge detection. This is done in both simulated scenarios and in scenarios involving actual physical smoke with synthetic smoke sources as the benchmark. I have shown in the presentation that the system achieves accuracy and F1-Score 85% and 75% respectively, where Model 3 that is trained with 80 percent data and tested on 20 percent data yields the best outcome

[41]. In addition, the use of Graphic Processing Unit (GPU) processing is employed, and it is 6.34 times more efficient than that of Central Processing Unit (CPU), which provides for real time operations, for the effectiveness of the system. It was trained using a small dataset containing only 280 smoke images which eventually affects its ability to recognize the smoke during fire incidences in different environments. With a broader set of images and higher number of such images, it is reasonable to note that the detection accuracy would also increase especially in different weather and lighting conditions. The system is right now designed to detect only the initial sign of a fire, which is smoke. However, it is unable to recognize fire or even smouldering fires and other burning contents that are already well developed. Incorporating features of flame detection could improve the system resolution and reaction time, thus making it more adaptive at different phases of fire growth. For instance, the system uses a Raspberry Pi for the processing needs at the same time, it may have the problem of computational constraints. It may affect the performance of the system in other detection tasks, such as day/night video surveillance, or in real-time processing of high-resolution images. To simulate the experiments of the system the synthetic smoke sources used which can't mimic some conditions which are possible in real world such as effect of wind on smoke dispersion or smoke from far away fires. The paper does not discuss the battery strength of the drone during surveillance especially when extended period is required in large, forested regions where surveillance is required in large areas.

2.1.12 Wildfire Detection from UAV Collected Images Using Transfer Learning

The paper proposed a video-based wildfire detection system using drones owning high-resolution cameras and transferring learning. It uses CNN pre-trained with wildfire images provided in the FLAME dataset, the dataset that includes 39,375 training images, and 8617 test images, all divided into fire and no fire. The study used transfer learning on CNN architecture including ResNet50, VGG16, VGG19, InceptionV3, and Xception for the binary wildfire detection tasks. Finally, the analysis of results showed that ResNet50 can be successfully used for fire classification and its accuracy was the highest in the experiment – 88% [42]. The system performed a preprocess of color normalization and data augmentations where the existing images were rotated and flipped to increase model resiliency. information on the real-time performance, low

values of the prediction times were confirmed, which makes system suitable for UAV-based Wildfire monitoring. The system proposals reliance on drones and transfer learning makes the system a cheap and efficient solution to previous more costly methods such as satellite imaging thereby meeting the requirement of early and efficient fire detection. The Thermal data or infrared data which is so crucial in detecting fires especially during heavy smokes or at night is missing in the FLAME dataset which is basically a set of RGB images. Perhaps, the lack of variation of environmental conditions might hamper generalization of the model during real fire incidences. While transfer learning expedites model creation, fine-tuning pre-trained models may not cover domain-shift issues, for example, between fires and similar-sounding phenomena, including sunsets or reflections. As it operates with high-resolution RGB images it can be rather limited by false negative – if the fire is hidden by smoke or vegetation the system will not notice it. Real-time computation capabilities may pose problems for UAVs because of the need to process large areas of footage. However, an important operation limitation is the battery energy capacity in drones and also the weather challenges that affect the battery energy source. The above analysis shows that the system has low robustness because it depends only on visual data. The integration of further data of thermal or hyperspectral imaging could enhance the performance in difficult detection situations. The model does not perform well where there are very few similar inputs as used in making the model as the general environment may be very different. More experiments on datasets which are different in terms of geographical location are required to increase the level of generalization of the retrieved results and the stability of the obtained algorithms in various conditions.

2.1.13 Applying Deep Learning to Real-Time UAV-Based Forest Monitoring

This paper presents an improved monitoring system for forests using UAVs equipped with RGB and infrared imaging sensors. The main contribution of this work is a 4-channel YOLOv5 object detection model that uses both RGB and thermal infrared images to enhance detection performance. Exploiting the complementary advantages of the two types of sensors, the system overcomes problems inherent to traditional models that rely solely on RGB data, like low light conditions, fog, and smoke interfering with visibility. The multi-sensor data captured by the UAVs are processed

in real time with the detection model, and results are displayed remotely via a web-based platform for monitoring purposes. An aerial dataset of thermal and visible images for forest surveillance was designed specifically for this purpose to ensure effective training and validation of the model under robust conditions. Results show substantial improvement in precision as well as Mean Average Precision (mAP) compared to state-of-the-art models under adverse conditions. Additionally, modular architecture allows scalability; thus some practical applications include real-time monitoring during wildfires prevention strategies, ecological management over timescales or space scales too vast for ground patrols, and forest surveillance [43]. The thermal and RGB imaging dependence makes the UAV hardware expensive and complex, thus potentially affecting its accessibility and scalability for general deployment, particularly in resource-limited environments. The reliance on high-quality, multi-sensor data can significantly reduce the system's effectiveness in areas with limited infrastructure for deploying UAVs or where connectivity is poor--conditions often encountered in remote forest regions. The tailored dataset enhances the accuracy of the model, but generalization is not at all assured to other forest environments with different terrains and climate conditions, thereby compromising the system's robustness in various regions. The current system is designed mainly to detect people and vehicles, ignoring other important monitoring features in the forest, like identifying smaller-scale changes in vegetation health or fire hotspots, which are crucial for integrated forest management. The requirement for real-time image transmission and processing results in substantial computational demands as well as the need for effective communication networks, which may prove difficult to sustain, particularly in remote or large-scale forested regions.

2.1.14 YOLO-Based Models for Smoke and Wildfire Detection in Ground and Aerial Images

This paper explores smoke and wildfire detection in terrestrial and aerial images using YOLO-based deep learning models. Moreover, the study encompasses challenges in detecting smoke and fire whose shapes, sizes, and colors changed and become complex with the background. YOLO architectures are machine learning architectures known for speed and accuracy; thus, it was a model of choice for real-time wildfire detection.

The study evaluated various YOLO variants such as YOLOv5, YOLOv7, YOLOv8, and YOLOv5n on two datasets: the D-Fire dataset, which consists of 21,527 images taken from different fire and smoke background scenarios as well as no fire to illustrate features not related to fire; and the Wildfire Smoke Detection from YouTube (WSDY) dataset, which contains smoke detection data with 737 annotated images [31].

Among them, it was found that the model YOLOv7x achieved the highest performance record for the D-Fire dataset, capturing an mAP of 80.40%, its robustness in detecting small size fire and smoke zones in complicated environments[31]. Meanwhile, for the WSDY dataset, YOLOv8s attained mAP of 98.10% and performed well in the early detection of smoke. This good performance is attributed to advanced feature pyramids, anchor-free detection, and data augmentation. Furthermore, lightweight models such as YOLOv8n and YOLOv5n will yield speeds without loss of power consumption, making them best suited for real-time applications. This proves that YOLO-based models have dominated the standard faster R-CNN methods in both accuracy and speed, thus proving their superiority in wildfire detection tasks. The quality and diversity of the training datasets have heavy influence on the performance of the models based on YOLO for wildfires detection. Databases such as D-Fire or WSDY may not cover all possible conditions possible during wildfires, thus leaving gaps in their detection capability once deployed in an unseen environment or region not represented in the data. Although models of YOLO like YOLOv7x and YOLOv8s provide high accuracy, they demand high computational resources, impossible to be deployed on edge devices with limited processing and power. Thus, they fail to yield real-time detections, especially in remote or resource-poor areas. False positives can occur when fog or clouds, which look similar visually, are classified as being fire related, making the system less reliant. But that can bring down the accuracy of the model during some scenarios, more so in complex environments containing a plethora of similar visual cues. This study does not talk about the long-term scaling capabilities of these models, under which they could be applied on the ground in large wildlife fires, or how these models would integrate with some other monitoring modalities such as multisensor systems or thermal imaging. Thus, it limits the potential of these developments for being broadly applicable to comprehensive wildlife fire monitoring solutions.

2.1.15 A Survey of Machine Learning Algorithms Based Forest Fires Prediction and Detection Systems

This paper presents a comprehensive review of ML approaches for predicting and detecting wildfire events, emphasizing the significant MST transformation that encumbers the dilemmas of wildfire. The first category includes fire prediction systems (before ignition and detection) while the second related category of fire detection systems deals with the identification and location of existing fire events. Among the diverse ML methods could be cited artificial neural networks (ANNs), which are very robust in cases of noisy data as well as produce up to 98.9% fire ignition prediction and detection accuracy [44]. Other referenced models include Logistic regression and decision trees, as these yield efficient models in fire ignition probabilities and susceptibility maps, thus providing reasonable estimates of environmental factors-fire risk relations. CNNs can be viewed as a significant breakthrough in DL models for image-based fire detection, with accuracies greater than 94% for detection of smoke and flame patterns [44]. Other algorithms, such as SVMs, Bayesian models, and Fuzzy Logic, among other hybrid models that exercise various models together, have produced even higher performance as well as robustness. The paper addresses also the implications of advanced technologies such as WSN and UAV convergence of ML. Prediction systems predict fire ignition through weather, fuel, and human activity along with the spread of fire. Detection systems utilize cameras, infrared sensors, and UAVs in real-time, as well as collect information from meteorological data, satellite imagery, and ground-based sensors for further investigation [44]. The major impediment in developing ML-based wildfire detection systems is the lack of large, high-quality datasets. The remote sites do not offer adequate or inconsistent data, thus restricting the generalization of the model and ultimately reducing the effectiveness of the system in different environments. Another aspect that makes real-time implementation difficult, even in resource-constrained environments such as remote forests, is the high computational power and energy demand of the models. The feasibility of ML-based solutions will be restricted even more in such areas where access to reliable communication networks cannot be possible. False positives and negatives are a major concern in wildfire detection, particularly when related phenomena must be

differentiated from mere visual analogs like fog, clouds, or industrial smoke. The models may also be inaccurate during sudden environmental changes like abrupt changes in the weather, causing prediction failures. Hybrid approaches, which use a combination of different models or techniques for attaining enhancement performance, can introduce additional complexity in the system and make deployment more expensive. Such added complexities must be fine-tuned to be cost-effective and practically deployable. The logistical considerations with large-scale deployment of ML-based wildfire systems over larger areas would include aspects like infrastructure maintenance, real-time data transmission, and the physical limitations of deployed sensors or UAVs. The present systems lack multi-sensor data fusion, like integrating thermal, Light Detection and Range (LiDAR) or multispectral imagery that hampers its proper functioning in a complex environment with varying conditions like smoke, low visibility or thick vegetation.

2.2 Existing Systems and Their Limitations

2.2.1 Traditional Fire Detection Systems

The conventional methods for forest fire detection, as mentioned in Fig 2.1, are in the form of manual observation, satellite imaging, and fire sensor ground-sensing systems, made to detect wildfires [45], [46]. However, these methods have few limitations, which restrict their efficiency, scale, and real-time detection capabilities.

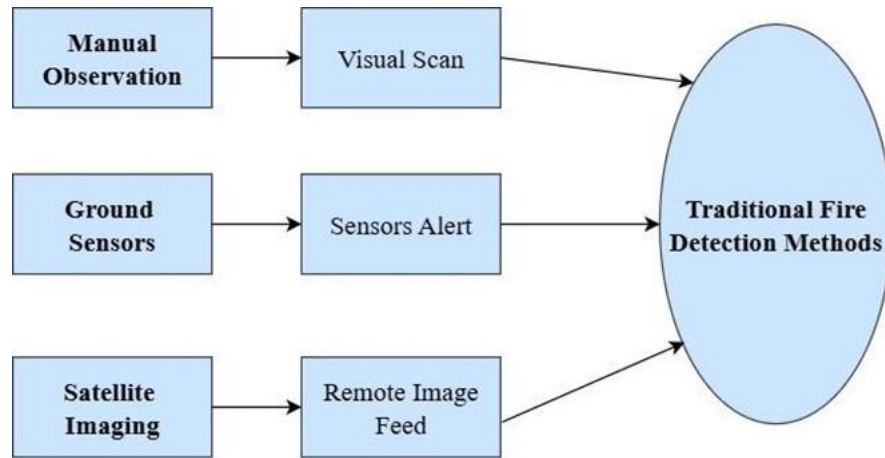


Fig 2.1: Traditional Fire Detection Methods

The use of fire watch towers and rangers to patrol the forest for fire scums is a traditional method but very labour-intensive and slow. This is highly human-error-prone, and it's also slow, particularly in large or remote forests, where manual coverage is limited.

The application of satellites covers broad areas, taking thermal and optical photographs of fire-risk locations. These systems of satellite suffer time delays during the transmission and processing of data, thus delaying fire detection. They can also be obscured due to cloud and other atmospheric conditions.

Ground-based-fire-detecting systems such as smoke and temperature sensors are also employed for monitoring the environmental parameters. These systems have a limited range of coverage, require a very frequent maintenance schedule to be followed, and are very prone to damage by environmental factors like wildlife or severe weather conditions.

2.2.2 UAV-Based Systems for Fire Detection

The introduction of UAVs or drones as fire detection systems, in other terms, is indeed defining something well beyond the capabilities of modern fire detection systems [43], [19]. As illustrated in Figure 2.2, the onboard thermal cameras can sense heat signatures and thermal anomalies to locate sources of fire. With the ever-increasing costs of thermal sensors, inertial effects such as smoke, fog, and even lighting complicate detection. It is possible to detect smoke in aerial photography by using drones that fly within the visible spectrum of light (RGB cameras) in the human eye. The near-complete coverage that weather conditions such as fog, low light, or bright sunlight can make mostly unavailable ends up affecting accuracy in detection. They have made it possible to provide real-time data and rapid surveys over very extensive areas

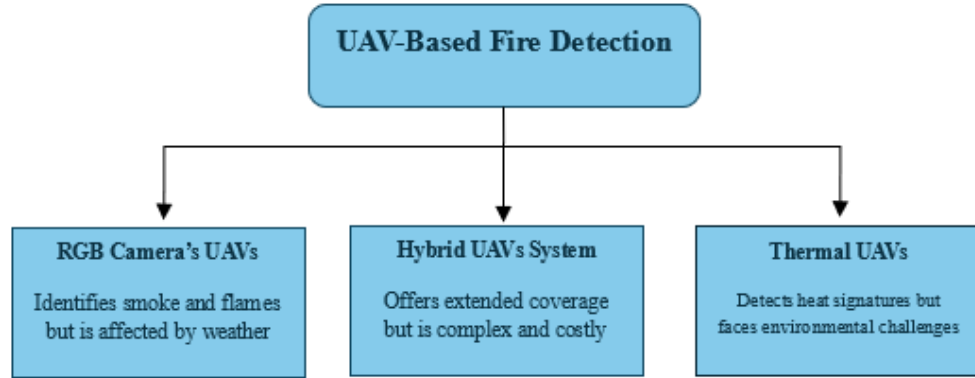


Fig 2.2: UAV Fire Detection System

In terms of coverage and time in the air, fixed-wing and rotary-wing drones make up an integrated system [47], [48]. Costlier, more complex, and difficult to operate efficiently than the systems mentioned above in Fig 2.2.

2.2.3 AI and Machine Learning Approaches: YOLO Models for Fire Detection

The methods for machine learning have contributed significantly to automating the fire detection process from images taken by UAV through their deep learning models, for example, CNNs [35]. One among these models is YOLO [20], specifically designed for real-time fire detection through its speed and accuracy. Over the years, the YOLO architecture has experienced evolutionary phases, whereby every version has shown improvement especially concerning the features related to the environmental complexities of detecting fire particularly smoke or flames. There is below a comparative table summarizing some key features, limitations, and applications from YOLOv1 up to YOLOv11 regarding fire detection.

2.3 YOLO Models

2.3.1 YOLOv1

This nucleation was the very first chapter of the YOLO series. In fact, it totally made a breakthrough in using the single regression way of treating object detection as a problem. That saw the whole image divided into grids as they would predict the bounding boxes plus class probabilities, all done at one go. Real-time object detection using a single convolutional network. Processing speed is higher than conventional

detection methods. Small objects made it difficult (example: small flames or smoke). Lower accuracy for complex environments where dense backgrounds are present. Resolution handling limits fire-related feature detection in large-sized images from UAVs. The application to fire detection was mostly in initial detection of fire in UAV images. However, its capabilities fell short of differentiating fire-related features, especially in cases where fire was small, or partially obscured within heavily forested environments [49], [50], [51].

2.3.2 YOLOv2

Also referred to as Darknet-19, YOLOv2 was the upgraded version of YOLOv1. The improvements of YOLOv2 over YOLOv1 included having a much stronger backbone and better treatment of small objects. Compared to YOLOv1, this one is higher in accuracy as well as higher in speed. Better for small-scale detection of objects, which is very important for the identification of small fires or small smoke plumes in UAV-captured images. Still facing challenges with false positives in cluttered backgrounds or complex fire scenarios. Poor generalization in dynamic fire environments with rapidly changing fire behavior. YOLOv2 helped improve real-time fire detection in UAV systems, particularly for smoke detection, and improvement of fire detection for small fires. Still, in complex outdoor environments such as forests, this system's accuracy was compromised by some false positives [50], [51].

2.3.3 YOLOv3

YOLOv3 is a modified version of YOLOv2 in such a way that it has increased depth and accuracy of the model. It also introduced an advanced architecture-darknet-53 that made it possible to detect a wider variety of objects with greater fidelity. Detection of small and large targets at different levels of resolution. Detection of very small objects like smoke and fire in different environments improved in recent times. Opposition to other models of its time in faster inference. Still prone to false alarms in cluttered or noisy environments, such as smoke that coalesces with cloud patterns. Although the fire spreads very fast, the fire features become hard to distinguish in such environments. The YOLOv3 was more efficient than other methods where small flames and smoke

were considered for UAV images in a complex fire scenario. With faster, more reliable results, the system also improved early detection of fires in forested areas [50], [51].

2.3.4 YOLOv4

The YOLOv4 object detection model has incorporated many optimizations, which improved speed and accuracy. In this regard, it serves as quite a multipurpose model for use in object detection tasks, including fire detection in remote areas. This has merged several new advances, such as the use of Mish activation functions and novel data augmentation techniques. Improved real-time detection handling in highly complex environments. Improved detection of small objects, which is quite crucial for fire detection using UAVs. Detection of fires by YOLOv4 is still a challenge in high-density smoke or fast-spreading fire environments. False positives in complex vegetation environments are still a concern. YOLOv4 has been used to develop state-of-the-art UAV fire detection systems that can detect fire features across greater areas and at faster speeds. Improved accuracy and efficiency would allow for more dependable detection of smoke and fire in a difficult outdoor environment [50], [51].

2.3.5 YOLOv5

YOLOv5, which is not a part of the YOLO family's official version, but it is popular in the fire detection community because of the sheer efficacy and flexibility it offers. Optimized for speed and easy deployment in real-time inference applications. Good enhancement in object detection, notably for small and remote fire phenomena such as smoke plumes. More flexible training and finetuning of the model for any specific fire-related detection task. Possibility of false positives in visually noisy environments, e.g. smoke or clouds. Very challenging to detect fires in large areas due to the lack of sufficient training data. YOLOv5 was one to frequently be applied in UAV-based fire detection systems. Even better with drone camera attachments to continuously monitor fire reactions, it became appropriate for wildfire detection due to its real-time detection of events associated with fire, like smoke [50], [51].

2.3.6 YOLOv6

Real-time applications in industrial environments have been enhanced with fire detection features using YOLOv6 via further optimizations. Better accuracy in detecting small objects like smoke or embers. Improved performance in complicated scenarios with dynamic movement and fast fires. Optimized for getting into edge devices, such as drones. False positives during fire-environment interactions with other similar environmental phenomena continue to occur. Extreme lighting or contrast situations are less effective, such as night fires. YOLOv6 has developed UAV fire detection, which enables real-time detection of fire-related features by drones in complex and dynamic forest ecosystems [50], [51].

2.3.7 YOLOv7

YOLOv7 provides the rapidly real-time object detection in highly dynamic environments, thus fitting the bill for wildfire monitoring. Better resistant to the effects of the environment like smoke and fog. Better multi-scale detection which is very vital for detecting fire features at various ranging distances. Flexible deployment options across the edge and cloud platforms. Large-scale fire scenarios still pose challenges, particularly those of rapid extension. Performance in detection can be influenced by extreme weather. YOLOv7 is more accurate and timelier in fire detection for capturing of the wildfire increase, in turn allowing intervention to occur earlier in larger forested areas [50], [51].

2.3.8 YOLOv8

Innovative features have been presented with YOLOv8 wherein it is stressed to have an increased efficiency for real-time applications in tough environments. Further improvements in small object detection, make it ideal for detecting smoke or small flames. Better handling of dynamic and changing environments, such as fast-moving wildfires. Optimized for real-time processing with low computation overhead. Causes false positives in low-contrast environments. There is still a challenge in the detection of fire in configurations of dense, smoky environments. YOLOv8 improved UAV fire detection systems to effectively improve the precision of fire detection in dense forests or intricate wildfire situations [50].

2.3.9 YOLOv9

Through modern enhancements on real-time object detection, in particular, smoke and small flames detection dedicated to detection of fire-related objects, YOLOv9 brings successful changes to such a task. Among its major improvements is the ability to perform better in the detection of low-contrast and small objects that are in most cases difficult to identify in forest and wildfire conditions. YOLOv9 pays attention to voxel-based traits and better feature mining to achieve heightened preciseness under problematic visual settings. Also, it is efficient in deployment on memory and computational power constrained devices like controllers, as it can perform real-time inference without accuracy loss. This renders YOLOv9 especially good at working with early-fire detection where little or no visual information can be visible due to smoke [51].

2.3.10 YOLOv10

YOLOv10 has improved on the success of its predecessor since it further enhanced the model in terms of detection under extreme environs like heavy smokes, bright lights or even quick spread of flame. YOLOv10 also promises to be a lot quicker and accurate when it comes to detecting any fire-related properties, such as shallow smoke to huge, speedy flames. It is field-programmable to enable real-time on edge devices processing, an aspect that is perfect in that aerial surveillance can be used in wildfire prone zones with the use of drones. YOLOv10 also features additional strong adaptation mechanisms capable of responding to difficult weather situations, which means that it is a powerful option to be used as part of an emergency response system where speed and stability are important factors in risky areas [51].

2.3.11 YOLOv11

If we talk about dynamic and challenging environments such as wildfire detection, we will probably want to talk about the new version in the series known as YOLOv11. Enhanced multi-scale detection for better detection of fire at various distances. On-feasibility against interference from the environment like smoke, clouds, and variations in light. Optimized for real-time applications with low latency - perfect for UAV-based systems. YOLOv11 is supposed to be the new standard for real-time fire detection

capabilities with both high precision and low false positive rates to deploy across large rural landscapes to detect wildfires [52].

2.4 Gaps in Existing Solutions

Yet, there are many gaps in the currently available fire detection technologies, despite the highly effective capabilities of YOLOv5, YOLOv8, and YOLOv11 for fire detection purposes. While YOLO models are quite strong, they may still generate false positives or detections because of noise in the environment. Although the work of YOLO in detection works perfectly, it generates false positives in detected areas where smoke and other disturbance elements make the scene look similar to fire. Another drawback is that, while offering a cloud-based processing mechanism such as that provided by Roboflow, it does not allow UAV systems using YOLO models to be very scalable in terms of the operational costs of drones, the computational requirements they need, and the complexity of operating and maintaining these large fleets across remote areas. Advanced hardware integration with YOLO based detection models in real-time can be problematic due to battery life and computational power requirements. Some of these gaps in fire detection are illustrated in Fig 2.3:

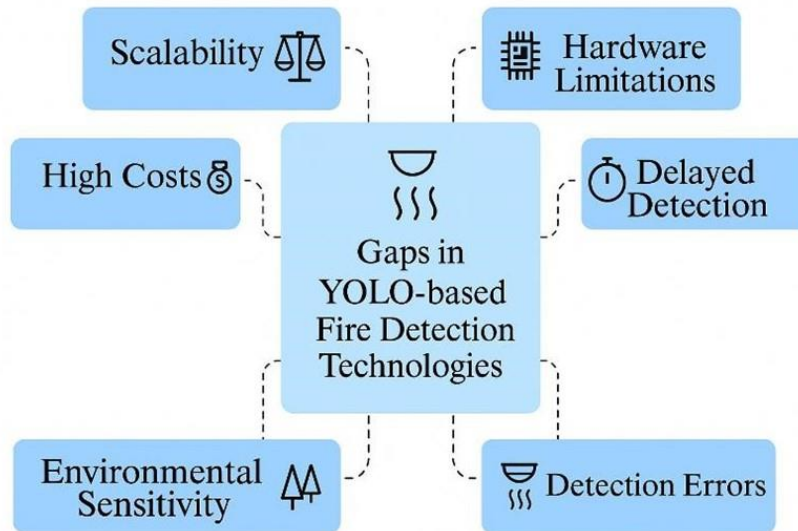


Fig 2.3: Gaps in Existing Solutions

While UAVs with YOLO models build lower-cost infrastructure when compared to satellites and thermal systems, hardware, such as drones and cameras, is still a

considerable cost barrier, especially for large-scale deployment in economically challenged areas. Even though YOLO models can detect events in real time, the processing time of UAV images before uploading them to cloud platforms such as Roboflow can still add a period of delay before they are interpreted, even more so for large-area monitoring. Smoke, fog, variable light, and other environmental reasons challenge fire detection based on both thermal and RGB imaging systems. While YOLO models are quite strong, they may still generate false positives or detections because of noise in the environment. Although drone-mounted systems offer flexibility, they still face hardware limitations related flight time, sensor accuracy, and computation power essential for real time detection [53].

2.4.1 YOLO Models Use Case in Fire Detection

Smaller drones can use YOLOv5n because it is lightweight, though it has some trouble processing detailed smoke. With YOLOv5s, you get improved accuracy for wider areas and moderate hardware. YOLOv8n can handle complicated environments more easily but requires additional computing. YOLOv8s is both quick and reliable for use on medium to large UAVs in varying conditions. YOLOv11n gives accurate results in problems with complex objects, but it won't work without the right hardware. YOLOv11s provides good results with not many false findings which is useful in large-scale situations [20], [51], [54].

Table 2.1: YOLO Models Use Case in Fire Detection

S.No	YOLO Model	mAP	Strengths	Limitations	Use Case in Fire Detection
1.	YOLOv5n	~80%	Lightweight, fast, real-time optimized	Struggles with dense smoke and extreme conditions	Best for small UAVs in less complex environments
2.	YOLOv5s	~85%	Fast, high accuracy, efficient	Requires moderate hardware for best performance	Suitable for real-time detection on larger areas
3.	YOLOv8n	~85%	Improved accuracy, fast inference	Higher computational demand than YOLOv5 models	Ideal for large-scale deployment in complex environments
4.	YOLOv8s	~88%	Better robustness, faster than previous versions	Slightly higher hardware requirements	Good for medium to large UAVs in diverse weather conditions
5.	YOLOv11n	~90%	Extremely accurate, robust to environmental noise	Requires high-end computational resources	Best for high-precision detection in challenging environments
6.	YOLOv11s	~92%	Top-tier accuracy, minimal false positives	Needs powerful hardware for optimal performance	Perfect for large-scale, high-accuracy fire detection systems

Various YOLO models for fire detection are compared in Table 2.1 based on their accuracy, key advantages and limitations.

CHAPTER 3

Design & Methodology

The objective of this project involves creating a forest fire detection system through image processing techniques and deep learning algorithms. The system detects fire together with smoke through analyzing live imagery because detectors have been replaced with this method to permit rapid responses and reduce large-scale forest destruction. To build the fire detection model we gather a dataset consisting of images with fire events and non-fire events. It collects forest imagery from Roboflow which guarantees the correct identification of flames and smoke areas. The model performance reaches higher levels through image preprocessing which includes resizing for standardization alongside normalization for consistent pixel values and data augmentation techniques of rotation, flipping and brightness adjustments to boost model robustness.

YOLO models including YOLOv5, YOLOv8 and YOLOv11s serve our fire detection process because these newest versions exhibit higher accuracy and processing speed than YOLOv3 and YOLOv4. The evaluation includes testing small scale versions of these models: YOLOv5n, YOLOv8n and YOLOv11n to determine their aptitude on low-resource hardware systems. We complete model training by refining yolov8s.pt through our dataset on the Google Colab platform. The detection system trains for 100 epochs to reach maximal performance while working with images of 640x640 pixels. The detection effectiveness of the model regarding fire and smoke is measured through precision, recall and mAP.

The model receives testing during real time within simulated software programs together with operational real-world settings after finishing its training process. The deployed best.pt model operates from a Raspberry Pi 4 for detecting fires in real-time. The system expansion includes creating a web-based platform which enables real-time fire detection tracking and reporting. Users can access this platform to get alerts as well as track fire locations using GPS data in addition to analyzing detection trends for better

forest fire management. Through image processing the system operates as a practical and scalable system for detecting early forest fires.

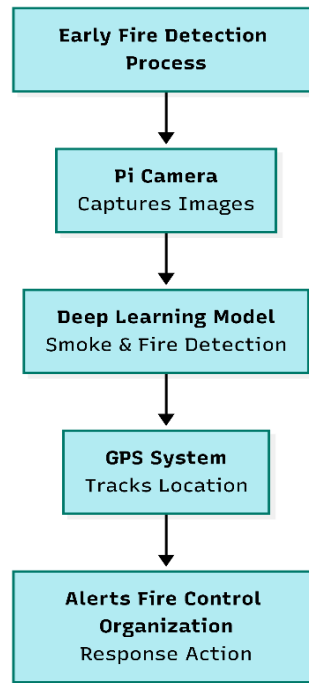


Fig 3.1: Early Fire Detection process

3.1 Dataset Preparation

The overall performance achievements of fire detection systems primarily depend on the dataset quality when both the training and testing processes take place. For successful discrimination, the model requires diverse samples to enable learning abilities that focus on real-world generalization. Roboflow functioned as the single platform to cleanse and refine the dataset, which used their specialized tools for managing and labeling, and augmenting datasets. The section describes the process used to obtain a data collection system that represents forest environmental conditions.

3.1.1 Dataset Collection

The public database delivers a dataset which contains information about diverse fire incident images. The database contains various essential categories that ensure precise fire detection along with smoke identification in forest areas. These classes include:

- **Fire Images:** The dataset encompasses images of spot fires together with images showing visible flames appearing in multiple forest areas.
- **Smoke Images:** The dataset contains smoke images that feature smoke bubbles together with loose smoke located near fire zones.
- **Smog Images:** The dataset contains images that depict thick atmospheric impurities such as haze and smog although these elements do not need to stem from fire-based sources.
- **Fog Images:** This class features images with blurry significance that natural fog produces because mist has similar characteristics to smoke.
- **Non-Fire Images:** Captures scenes without fire, smoke, smog, or fog to serve as negative samples.

False positive identification diminishes because the model applies data that lets it recognize fire-related indicators versus other non-fire elements. The distribution of the dataset appears in Table 3.1 shown below:

Table 3.1: Table of Classes and their Division into Training and Testing

S.No	Category	Images per class
1.	Bonfire	376
2.	Cloud	580
3.	Fire	845
4.	Fog	650
5.	No Fire	377
6.	Smoke	672
7.	Total	3500

3.1.2 Annotation

The process of annotation magnifies unprocessed images into trainable data components which brings true value to AI applications. The annotation process through Roboflow's tools includes manual review and labelling of each image that undergoes the analysis:

- **Smart Polygon Annotation:** The system uses smart polygon annotation methods instead of usual bounding boxes for detailed marking of irregular shapes of fire, smoke, smog and fog compounds.
- **Class Labels:** Annotated areas receive specific category labels which correspond to their class types either as "fire" or "smoke."

The annotation process focuses on handling difficult examples, including the combination of smoke, fog, and weak fire detections under low light conditions. Detailed annotation methods enable the model to distinguish minimal variations that exist between different categories.

3.1.3 Data Augmentation

Data augmentation techniques strengthen the model's resistance to real-world data challenges:

- **Flipping:** The model benefits from different angles with horizontal-vertical flips of images.
- **Rotation:** Throughout the process, data augmentation performs rotation by applying different image angle transformations.
- **Brightness Adjustments:** The model accepts adjustments to image brightness through a mechanism that resembles different lighting environments, such as dawn, dusk, and overcast days.
- **Blurring:** The method of blurring simulates images with motion blur or poor camera quality, which frequently appears during forest monitoring.

The augmentations expand the dataset beyond its original scope while broadening its variability, which results in better generality for unknown inputs.

3.1.4 Dataset Splitting

Three subsets exist to facilitate proper model training along with testing and evaluation processes:

- **Training Set (70%):** The Training Set (70%) contains most images for model training purposes to enable pattern recognition across each class.

- **Validation Set (20%):** The Validation Set (20%) operates independently to optimize model training through fine-tuning because it helps prevent the phenomenon of overfitting.
- **Testing Set (10%):** The Testing Set constitutes 10% of the total images which the model uses to demonstrate its ability to generalize on unpublished pictures for real-world applications.

A robust fire detection model can be trained with the help of a reliable foundation provided by a dataset containing multiple classes, including fire, smoke, cloud, fog, bonfire and no fire-smoke, with a complete annotation and augmentation implementation. The diverse collection of images provides the system with the capability to detect fire indicators effectively while reducing potential non-fire false positive alerts.

3.2 Model Selection and Training for Early Fire Detection

Success of the early fire detection system depends on making a proper model choice. We focused on selecting a model which would balance accuracy with efficiency for working with expansive forest images since we used large-scale forest images. The following section will discuss the model selection steps and training procedures.

3.2.1 Data Preparation and Dataset Utilization

A selection of images from Roboflow was used to train the fire system, showing fire and smoke in different forest environments.

All the images in the data were annotated to help tell apart fire, smoke, smog and fog. This allowed the model to understand the visual properties needed to detect early fire events. Since the data included various situations, the system became better suited for working with actual forest environments.

After training, the detection model was tried out with a Raspberry Pi 4 connected to a Pi Camera Module. The system was able to receive video/image from a drone in real time and respond to early fires in challenging areas.

3.2.2 Model Evaluation

YOLOv5 appears to offer a strong compromise between quick detection, high accuracy, and efficient use of resources, because YOLOv5n is so small and fast, it works great on limited-power systems, making it perfect for small drones. Alternatively, YOLOv5s gives slightly better results, but takes less time to process, fitting the needs of big coverage and decent computing resources. Both kinds of detectors work well in discovering smoke and early fire and provide good results in a reasonable period. These models are efficient enough under tight conditions to be used in mobile and embedded fire surveillance systems.

3.2.3 Training Process

Unoptimized deep learning training took place through an object detection framework optimized for real-time scanning. The Ultralytics framework using PyTorch stood as the training option because it delivered easy operation combined with flexible model management through support for YOLO versions YOLOv5 and YOLOv8. The deployment and training via validation steps of this framework enables easy use for fire detection platforms.

Training Parameters

- **Batch Size:** We selected batch size 16 as it achieved a balance between computation speed and training stability.
- **Epochs:** The network executed 100 epochs to build sufficient understanding from various fire along with smoke conditions.
- **Learning Rate:** The learning rate value was set to 0.001 for achieving steady convergence of the training process.
- **Loss Function:** The loss function improved both detection precision by optimizing bounding box outputs and reducing unneeded false alarms.

Evaluation Metrics

Evaluation of the model performance through important metrics revealed its performance assessment after completion of training:

- **Precision:** The detection accuracy of the model represents how precisely it locates fire and smoke occurrences while controlling unnecessary alarms.
- **Recall:** Counting all fire and smoke occurrences is evaluated to confirm the model detects all possible incidents of early detection.
- **Mean Average Precision (mAP):** The detection performance assessment provides an overall performance metric through mAP which determines detection accuracy across multiple detection conditions.

3.2.4 Model Compilation

Building the early fire detection system started with the important step of compiling the model. During this step, the team optimized the model to work efficiently with the hardware and to process all input data in real time. We wanted to improve the model so it could accurately find fire and smoke and respond quickly enough for use in forests.

Using a deep learning framework that can do dynamic computations and deploy easily, the model was developed and trained. The learning rate, batch size and number of epochs were all adjusted during training to achieve an optimal solution with both high accuracy and reasonable efficiency. It was important to the training that the drones knew how to distinguish between fire, smoke, fog and similar sights so as to limit false alarms.

When the model was complete, it was compiled and made efficient for deployment with a Raspberry Pi 4 and a Pi Camera Module connected. Since the hardware was limited in resources, we trimmed and quantized the model to make it smaller, quicker to run and still accurate for detection. Because of this, real-time applications with urgent decision-making needs could use the system.

The performance of the compiled model was tested using precision, recall and mAP. With these measurements, the model performed well in finding fire and smoke while avoiding errors in detection. All these configurations were tested for latency to confirm that the system was quick enough to help detect fire incidents in a timely way.

The model was placed into the system, so it could examine any images caught by the Pi Camera for fire and smoke in real time. As soon as a detection took place, the system transmitted the results, the image and the location information to the web portal.

3.2.5 Testing and Validation

During the testing and validation phase, the trained model was evaluated to ensure reliable performance on new, unseen data. For testing, a set of randomly sourced forest images entirely separate from the training dataset was used. These images were selected to represent real-world scenarios, including various lighting and weather conditions, as well as diverse backgrounds. The model was assessed for its ability to detect fire and smoke accurately while avoiding false positives caused by similar elements such as fog. Since none of the testing images were part of the training dataset, this helped validate the model's generalization capability and minimized the risk of overfitting. The results showed that the model consistently identified fire-related events with high accuracy and minimal false alarms, supporting its readiness for real-time deployment in early forest fire detection systems.

3.3 Software Implementation

3.3.1 Purpose of Raspberry Pi Imager

Raspberry Pi Imager performs the vital role of installing operating systems onto microSD cards for enabling the boot functionality of Raspberry Pi 4. Through its user-friendly platform the tool lets users choose Raspberry Pi models and operating systems while setting up storage devices for installation. The operating system installation workflow becomes easier so the Raspberry Pi can deploy without requiring human assistance.

3.3.2 Significance in Fire Detection System

The Raspberry Pi 4, powered by Raspberry Pi OS, serves as the central processing unit in the real-time fire detection system. Its role is critical and can be highlighted as follows:

- **Stable Operating Environment:** Raspberry Pi OS offers a lightweight, stable, and Linux based platform, ensuring efficient execution of deep learning models and image processing algorithms.
- **Library and Framework Support:** It supports seamless installation of essential libraries such as PyTorch for running deep learning models (including YOLO-based fire detection) and OpenCV for real-time image and video processing. The YOLO model enables accurate detection of fire and smoke in captured frames.
- **Hardware Integration:** The Raspberry Pi 4 ensures compatibility with essential hardware components, including the Camera Module for image acquisition and the GPS Module for tagging fire locations with geolocation data. For wireless communication, the system utilizes the Raspberry Pi 4's built-in Wi-Fi module, enabling real-time transmission of detection alerts and GPS coordinates to web portal, eliminating the need for an external communication microcontroller.

By leveraging the Raspberry Pi 4, the fire detection system achieves low power consumption, cost-effectiveness, and real-time performance, making it a reliable and scalable solution for early fire detection.

3.4 Hardware Specifications

The early fire detection system employed hardware components which guarantee efficient real-time reliable fire detection capabilities in forest environments. The hardware system consists of indispensable components which function together to collect information before transforming it into usable data for authorities. The following section includes an in-depth outline of the utilized hardware components.

The selection of hardware components in this system achieved efficient reliable time-sensitive fire detections in forest conditions. Every hardware component serves essential functions to gather data and transform it into useful information directed at authorities. The following section provides detailed information about the implemented hardware system:

3.4.1 Raspberry pi-4

The Raspberry Pi 4 shown in Fig 3.2 serves as the core processing unit in our early fire detection system [55]. The Raspberry Pi 4 unit manages image data through an AI processing system to execute deep learning models while connecting and exchanging data between the GPS module and web server through its standard dual-band Wi-Fi. The selection of Raspberry Pi 4 happens because this device mixes cost-effectiveness with mobility features and supports execution of lightweight AI models perfectly suited for edge computing in inaccessible locations.



Fig 3.2: Raspberry Pi 4 [55]

Hardware Setup

The hardware configuration includes the following key components:

- **Raspberry Pi 4:** The model execution of YOLOv5s fire detection takes place within the central computing unit.
- **Camera Module:** Real-time video frames are captured as an integral part of anomaly detection for fire, smoke and associated abnormalities.
- **Ublox NEO-6M V2 GPS Module:** Fire localization receives support from precise geospatial data provided by the software.
- **Built-in Wi-Fi Module:** The detection system connects wirelessly to a remote web portal through which it transmits results alongside coordinates.
- **Power Supply and Connectivity:** Operation stability comes with an additional battery backup system that enhances outdoor deployment capabilities.

The CSI interface enables a connection between the camera module and the Raspberry Pi and UART serial communication maintains the connection between the GPS module.

Software Implementation

Raspberry Pi runs a specific collection of software programs called the software stack:

- **Operating System:** The selected operating system is Raspberry Pi OS (Lite) because it runs light and supports headless setups.
- **Deep Learning Frameworks:** YOLOv5s inference operation, along with real-time video processing functions, are managed by OpenCV and PyTorch.
- **Custom Python Scripts:** This system was created to collect images using GPS data processing and fire detection functions that generate alerts.
- **Communication Protocols:** The system uses a Flask server for transferring fire detection data over Wi-Fi to a central web portal and monitoring system. The Flask server handles real-time communication between the fire detection module and the web interface by processing and serving detection alerts. This setup ensures efficient and reliable data transfer, enabling immediate monitoring and timely response in case of fire incidents.

The YOLOv5s model receives TensorRT and Open Neural Network Exchange (ONNX) optimization allowing fast and efficient ARM processor-based inference on Raspberry Pi devices. Early warning signals of fire as well as smoke detection are possible with this system.

Performance and Challenges:

Research measured the Raspberry Pi using detection accuracy and power efficiency as well as inference speed. Observations include:

- **Inference Speed:** The YOLOv5s model functions at speeds between 2 and 3 frames per second (FPS) which satisfies needs for real-time fire detection.

- **Accuracy:** The system performs very well at detecting both fire signs and smoke occurrences, but produces occasional wrong alerts during foggy situations.
- **Power Management:** Operation reliability requires a stable power source thus a backup battery system was added for continuous operation.

3.4.2 Camera Module

The Raspberry Pi Camera Module V2 shown in Fig 3.3 provides users with better image quality at a higher resolution than the original Raspberry Pi camera model. The Sony IMX219 sensor integrates into the device to give outstanding results when using computer vision systems for fire and smoke detection.

Characteristics:

The system uses the Raspberry Pi Camera Module V2 as its official camera component for Raspberry Pi boards [56]. The Sony IMX219 image sensor functions as a CMOS chip in the module to produce static images with 8-megapixel high definition at 3280×2464 pixels resolution. The Raspberry Pi Camera Module V2 provides multiple video recording modes starting from 1080p at 30 frames per second (fps) up to 720p at 60 fps and 640×480 at both 60 and 90 fps which helps users optimize between image clarity and computational capacity. Through its Camera Serial Interface (CSI) special port the module connects to Raspberry Pi while ensuring fast image reception that is critical for real-time fire smoke detection systems. Due to its compact size and excellent imaging features this module functions properly in embedded vision applications which need to operate in dynamic outdoor situations.



Fig 3.3: Raspberry Pi Camera [56]

Role in Fire Detection System:

- Captures real-time images and video for fire and smoke detection.
- Works seamlessly with YOLOv5s for real-time inference on Raspberry Pi.
- The device operates efficiently with low power requirements, which is ideal for drone-based uses.
- Higher resolution in the sensor leads to precise fire and smoke detection capabilities over different distances.

Connection with Raspberry Pi:

The camera module establishes a connection through its ribbon cable to the CSI port located on the Raspberry Pi 4. Through the libcamera or Picamera2 library the Raspberry Pi achieves efficient control over the camera system for image acquisition and processing tasks.

3.4.3 UBLOX NEO-6M V2 GPS Module for Raspberry Pi

The Ublox NEO-6M V2 GPS Module shown in Fig 3.4 serves as a compact reliable positioning device in fire detection systems to find out exact location details of detected hazards. This device provides fast geolocation capabilities to tag events related to fire and smoke occurrences in remote locations.

Specifications:

The Ublox NEO-6M V2 GPS Module contains the Ublox NEO-6M chipset which provides reliable positioning accuracy. The NEO-6M V2 GPS Module reaches location accuracy levels of 2.5 meters which makes it appropriate for independent fire alarm systems in open areas. Auditory data transmission between the Raspberry Pi and the module functions through the UART (serial) interface that delivers standard NMEA output protocols for location information exchange. Users can rely on the included ceramic antenna for secure satellite signal performance because it eliminates the requirement for external antennas. The module works using power supplies from 3.3V

to 5V while providing a default update rate of 1 Hz which delivers continuous real-time location information.

Use in Fire Detection System:

Through UART connection the GPS module sends continuous updates to the Raspberry Pi 4 [57]. Real-time alerts transfer to a remote web server through Python scripts which extract both latitude and longitude data together with detection results. The integration enables precise determination of fire locations to help emergency responders develop timely emergency plans.



Fig 3.4: The Ublox NEO-6M V2 GPS [57]

3.5 Fire Detection Web Portal

Web portal acts as a unified interface for supporting time-sensitive fire observation and detection visualization alongside emergency management coordination. The web portal functions through an architectural design that combines a Raspberry Pi-based drone with fire detection AI and Firebase backend functionalities. It enables smooth communication between recorded field data and fire response teams for instant data sharing for better decision-making.

3.5.1 Real-Time Detection and Visualization

The portal's fundamental capability involves showing fire detection results as they occur. Drone operations utilize the Raspberry Pi camera to acquire images, which the onboard lightweight YOLOv5s model analyses on the spot. Appropriate metadata, along with detection probability and severity level, is transmitted to Firebase Realtime Database or Firestore following fire or smoke detection. The web portal achieves real-

time alerts on the user dashboard through persistent data fetches enabled by its established listeners.

3.5.2 Drone Surveillance and Image Feed

The WiFi module of the Raspberry Pi allows uploaded aerial images/video to be sent to Firebase Storage. The web portal starts showing images automatically after detecting fire and smoke incidents to let users verify alerts and check where the fire has reached. The system improves operational awareness while assisting on-site personnel to coordinate their activities.

3.5.3 GPS-Enabled Fire Location Tracking

The spatial awareness capability of the Raspberry Pi system runs continuously as it gathers GPS tracking data that it sends to Firebase. The web portal utilizes Leaflet.js to display real-time mapping of data based on coordinates that come from the system through the open-source JavaScript library. Detection points include all necessary details such as their locational data and timing. The system enables users to track incident distribution across geographical areas and make efficient decision about resource allocation.

3.5.4 Historical Logs and Analytics

Each detection event recorded by the system writes information about time alongside location as well as model confidence levels to Firebase Firestore. The web portal applies Chart.js library to transform this database information into user-friendly visual reports. The picture-based elements enable users to detect high-risk zones as well as measure system performance throughout time.

3.5.5 Alert Notification System

The portal features a system that automatically generates alerts for quick responses. When a fire event occurs, the portal automatically sends out immediate email notifications and push alerts through Firebase Cloud Messaging. The portal sends immediate notifications to users who have selected roles and attributes based on their requirements to decrease response delays.

3.5.6 User Authentication and Role-Based Access

The system enlists Firebase Authentication to conduct both security and access control functionalities. Secure electronic password and email procedures enable user registration and authorization into the system. The system implements role-based access controls such as admin, operator, and viewer, which limits sensitive data and specific features only to authorized personnel. The system enables optional implementation of multi-factor authentication (MFA) as an additional security measure.

3.5.7 Centralized Dashboard and Interface

The portal uses React.js as its development framework, together with Tailwind CSS for styling, while maintaining a responsive format that works across different devices. The web portal unites various core features through a single dashboard, which brings together fire alert visibility and detection aggregate views with camera visuals alongside GPS fire visualizations and analytical system logs. The streamlined dashboard through this interface improves decision-making performance and enables better inter-stakeholder coordination for fire monitoring and emergency response.

3.5.8 Firebase-Based Data Infrastructure

Through Firebase services, the system operates its backend functions using scalable and reliable features that enable real-time data synchronization. Several essential Firebase elements power the portal to deliver an efficient operation. The system uses Firestore with the Realtime Database as storage systems to track fire detection outcomes and GPS data points, and stores image data through Firebase Storage. The portal uses Firebase Authentication as its authentication system to provide role-based access control in addition to robust secure login capabilities. The system achieves reliable high availability and consistent performance levels through Firebase's strong cloud infrastructure under changing and intensive operations.

3.6 Detection Workflow

The installed camera modules from field units performs forest environment monitoring by capturing images. The YOLOv5s deep learning model conducts image processing

at the field units which detects fire and smoke visual indicators. When a system detects anomalies, it performs visual highlights through box annotations while applying severity tags to the affected areas.

The detection results and their GPS coordinates, together with timestamps, are automatically transmitted to the database. The website displays all results so users can interpret and respond instantly. The system provides fast and organized firefighting through its automatic sharing mechanism of detection alerts to responsible authorities.

3.7 GPS Integration for Location Tracking in Fire Detection

The system integrates GPS technology to track the exact location of detected fires. When the YOLOv5s model identifies fire or smoke in an image, the GPS coordinates are captured and sent to authorities for immediate response as shown in Fig 3.5 below:

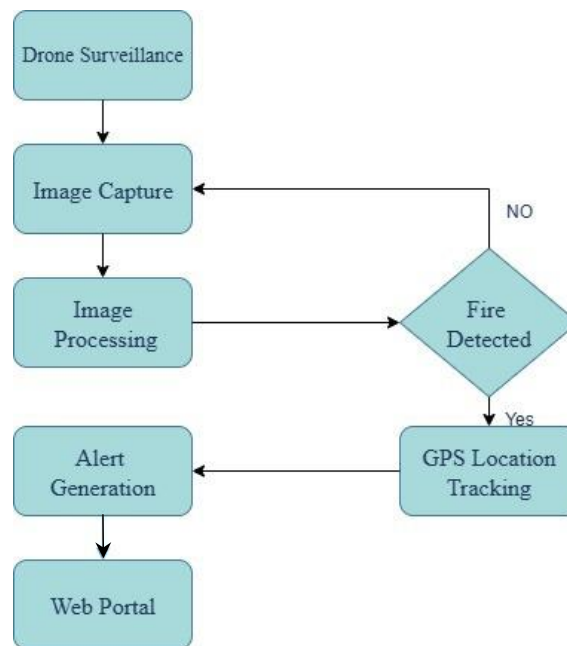


Fig 3.5: GPS Integration

The GPS data ensures that firefighting teams can assess the severity and respond effectively to mitigate risks and reduce response time.

3.8 Technologies and Tools Used

Early fire detection depends on a complete collection of hardware and software technologies which the system deploys for reliable and efficient operation. YOLOv5, YOLOv8 and YOLOv11 deep learning models work together for fire and smoke detection to achieve precise results in varied environmental contexts. Next.js together with React.js powers the web portal front-end and Tailwind CSS provides the system with a contemporary UI that delivers effective interaction between users and the interface. Node.js together with Firebase operates on backend operations to provide scalable storage features and authentication services and alert functions. The application includes Google Maps API through which users receive real-time representations of detected fire sites that are traced using GPS positioning data. When situated at the edge position the Raspberry Pi 4 executes program instructions on collected camera data while performing AI task computations directly at the device level. It later sends results via wireless transmission. The system uses Firebase Firestore and Firebase Storage to manage real-time data storage of metadata.

CHAPTER 4

Results and Implementation

The chapter details the complete execution and evaluation procedures for the developed Early Fire Detection System shown in Fig 4.1. FireWatch is the name given to the system which detects forest fires and smoke by processing aerial images through deep learning frameworks. The chapter describes the utilized hardware equipment together with software tools while explaining dataset annotation procedures and YOLO model training and assessment followed by Raspberry Pi deployment and FireGuard web portal development through its frontend and backend implementation.



Fig 4.1: Early Fire Detection in Forest using Image Processing Drone

4.1 Tools and Libraries

A compact embedded system installed on a drone platform served to execute the program. The central processing unit comprises Raspberry Pi 4B to execute real-time image acquisition functions and perform model inference and data transfer operations.

A high-resolution aerial imaging function was obtained from connecting a Pi Camera Module to the Raspberry Pi. The system employed Ublox NEO-6M V2 GPS Module to precisely track both the latitude and longitude positions of detected fire or smoke occurrences. The built-in Wi-Fi module of the Raspberry Pi transferred real-time images and coordinates which the system captured through its wireless transmission method. The Raspberry Pi 4B operated as the central processing unit and its Raspberry Pi Operating System (OS) was designed to work with the built-in Wi-Fi module along with being energy efficient. A drone frame mounted all hardware components which enabled the system to have both aerial capabilities and mobile functionality.

The system implementation included powerful libraries alongside software tools. The implementation of system logic used Python as its main programming language. OpenCV served to process images and depict bounding boxes with real-time video frame management capabilities. PyTorch implementation with YOLO models including YOLOv5, YOLOv8, and YOLOv11 operated for deep learning object detection of fire and smoke. The training procedures for all models utilized Google Colab that enabled Tesla T4 GPU acceleration. The development of the FireGuard web portal backend used the Flask framework as its base technology. The developers selected Next.js and React.js as their main technologies to build dynamic interfaces that users could interact with. TypeScript was incorporated to both increase type security and eliminate runtime mistakes while developing the system. The platform used Firebase for authentication management as well as real-time data synchronization throughout all elements.

4.2 Dataset Preparation and Annotation

The training models received their data from the combination of publicly available datasets with drone-captured aerial images that were specifically collected for the purpose [58], [59], [60]. The annotation platform Roboflow enabled the professionals to perform the dataset annotation in the Roboflow cloud platform.



Fig 4.2: Showing Class Bonfire and Smoke



Fig 4.3: Showing Class Smoke and Fog



Fig 4.4: Showing class Fire and Smoke

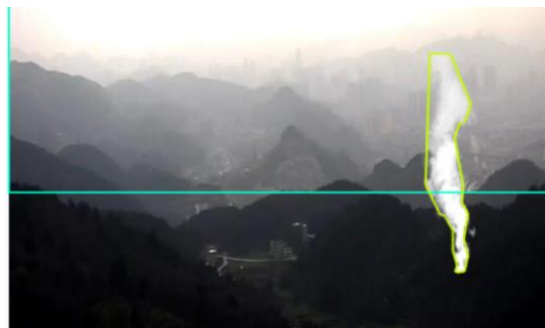


Fig 4.5: Showing class Smoke and Fog

Fig 4.2, 4.3, 4.4 and 4.5 screenshot from Roboflow showing annotated regions for fire, smoke and fog using bounding boxes [61].

The 3,500 images in the complete dataset were distributed into 70% training images (2,445 images) and 20% validation images (710 images) along with 10% testing images (345 images). The collection consisted of six different picture categories which included fire, smoke, bonfire, fog, cloud and no fire-smoke. A uniform resolution of 640 by 640 pixels was applied to all images while auto-orientation was used to correct alignment and the dataset underwent augmentation through rotation and flipping and brightness adjustments for improved generalization and robustness.

4.3 Model Training and Evaluation

A group of YOLO models received training on annotated dataset within the Google Colab environment that utilized Tesla T4 GPU. The research aimed to select both an accurate and energy-efficient detection system between the models for fire and smoke recognition. The models trained included YOLOv5n, YOLOv5s, YOLOv8n, YOLOv8s, YOLOv11n, and YOLOv11s. Every model received identical training parameters which involved running 100 SGD optimizer epochs along with a 16 batch size. The evaluation metrics included mean Average Precision (mAP) as well as precision and recall metrics together with a measure of overall classification accuracy. All models underwent performance comparison to establish the optimal selection based on their deployment practicality and operational outcomes.

Table 4.1: YOLO Models Performance Summary Over 100 Epochs

S.No	Model	Epochs	Precision (P)	Recall (R)	Accuracy (mAP@0.5)
1.	YOLO v5n	100	0.778	0.768	0.819
2.	YOLO v5s	100	0.747	0.786	0.815
3.	YOLO v8n	100	0.74	0.788	0.812
4.	YOLO v8s	100	0.784	0.764	0.815
5.	YOLO v11n	100	0.772	0.771	0.817
6.	YOLO v11s	100	0.771	0.785	0.812

As shown in Table 4.1 YOLOv5n delivered the best accuracy performance at 81.9% among the selection of available models. Testing and evaluation led the team to choose YOLOv5s as the model to deploy on the Raspberry Pi platform. YOLOv5s proved to

be the better choice for the Raspberry Pi 4B deployment due to its faster inference speeds along with greater stability and better resource utilization despite having an accuracy level of 81.2%. Even though YOLOv5n achieved higher accuracy results it delivered variable bounding box predictions that needed tuning in order for the system to function efficiently on low-resource devices. YOLOv5s operated with constant detection stability as it processed images in 1.5 to 2.5 seconds on average. Its performance capability made the system ideal for embedded fire detection systems which have limited computing capabilities.

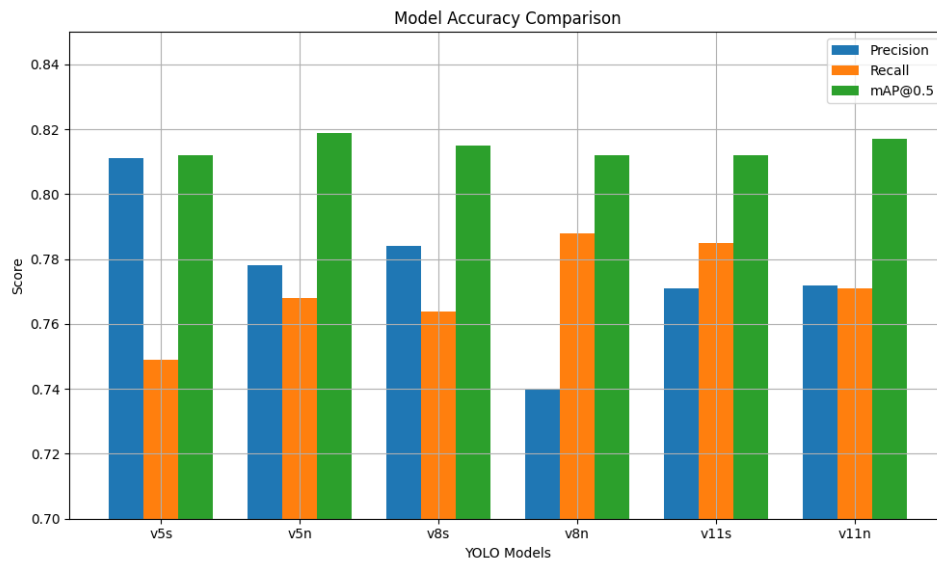


Fig 4.6: Comparative analysis of YOLO models

Deployment of YOLOv5s proceeded due to its ability to achieve accurate performance and efficient speed capability. In Fig 4.6 a graphical depiction show precision, recall, and accuracy data during 100 training epochs exists for evaluating each YOLO model's performance. The visual chart sums up the trade-offs between different model variants and displays the performance advantages as well as limitations of each architecture structure.

4.3.1 Training Progress of YOLOv5s

The performance evaluation of YOLOv5s throughout 100 training epochs appears in the Fig 4.7 graph. The YOLOv5s model demonstrated the best performance stability along with a low imbalance which made it ideal for embedded systems on Raspberry Pi devices. The performance of the model in detecting fire and smoke features is

measured through Precision and Recall alongside the mean Average Precision at 0.5 IoU ($\text{mAP}@0.5$) plotted metrics.

The model demonstrates its ability to avoid misidentification through precision and simultaneously demonstrates its capacity to identify all genuine fire or smoke cases through recall and delivers comprehensive localization and classification results through $\text{mAP}@0.5$. Analyzing the data from the graph reveals that YOLOv5s maintained consistent enhancement of Precision and Recall and $\text{mAP}@0.5$ while avoiding sudden changes. The model demonstrates proved generalization properties and prevents overfitting which makes it applicable for aerial detection operations with minimal computing resources.

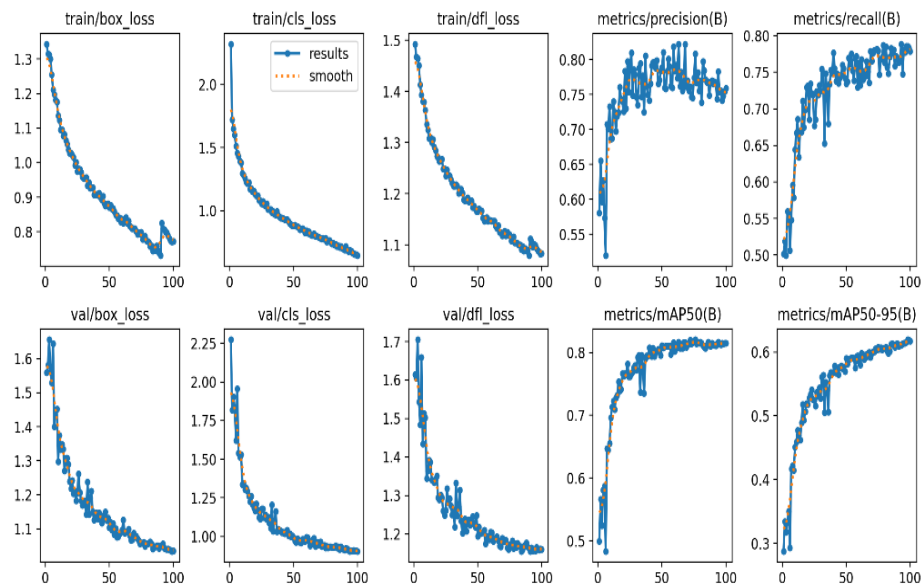


Fig 4.7: Training Performance of YOLOv5s over 100 epochs

4.4 Detection Performance on Static Images

The trained YOLO models underwent evaluation through the use of fresh new images which did not contribute to their original training data. Multiple images were selected for testing purposes because they displayed various actual fire situations with smoke present. The testing objective focused on model reactions to unseen visuals because real-world deployment situations involve such scenarios. The examination of models took place on Google Colab for standardized execution of tests.

The examination included running six YOLO models YOLOv5s, YOLOv5n, YOLOv8s, YOLOv8n, YOLOv11s, and YOLOv11n under three detection confidence settings of 0.15, 0.5, and 0.7. This parameter controls when a model decides to detect an object by defining its confidence levels and lower thresholds enhance sensitivity while higher ones optimize precision.

4.4.1 Results Overview Across Thresholds

At Threshold 0.15

The models identified increased numbers of fire and smoke incidents when using the 0.15 threshold because the lower confidence criterion allowed them to detect more obscure signals. The models detected delicate fire and smoke indications at the 0.15 threshold, but this detection approach heightened the possibility of incorrect positive results. Among all models YOLOv8n achieved the most fire detections numbering at 9 while YOLOv8s trailed behind with 8 detections. Though YOLOv5s detected fewer instances it performed outstandingly by detecting five fire instances and one smoke instance with accurate bounding box determination shown in Fig 4.8. YOLOv5s proved to be the most stable model because it delivered accurate result interpretation with minimal detection noise when recognizing essential features.

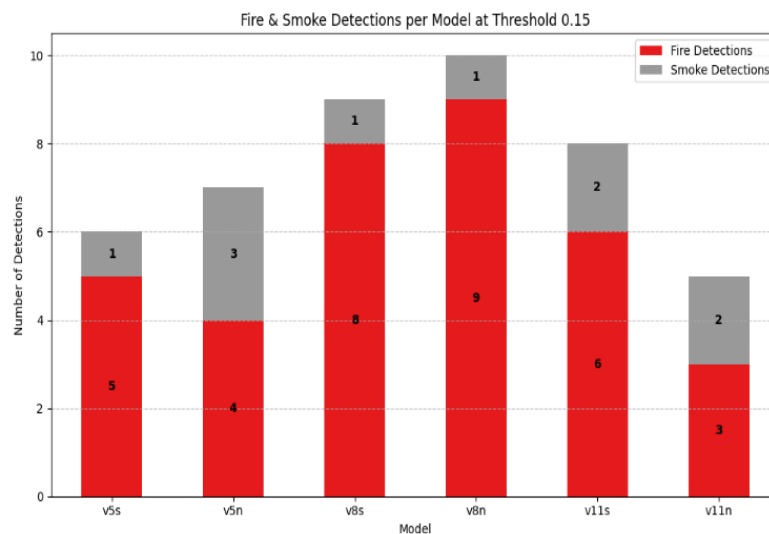


Fig 4.8: Fire & Smoke Detections per Model at Threshold 0.15

At Threshold 0.5

Higher confidence threshold settings at 0.5 led to precise and selective detections because the models dropped all lower-confidence outputs. All model detection counts shrunk because of the elevated confidence threshold of 0.5. YOLOv8s detected three fire objects to become the top performer contrary to other models regarding fire detection at moderate confidence levels. The combination of YOLOv5s YOLOv5n YOLOv8n provided reliable two-class detection by observing both fires and smoke automatically under intensified confidence restrictions as shown in Fig 4.9. All models maintained their detection capabilities when the threshold measurement was raised.

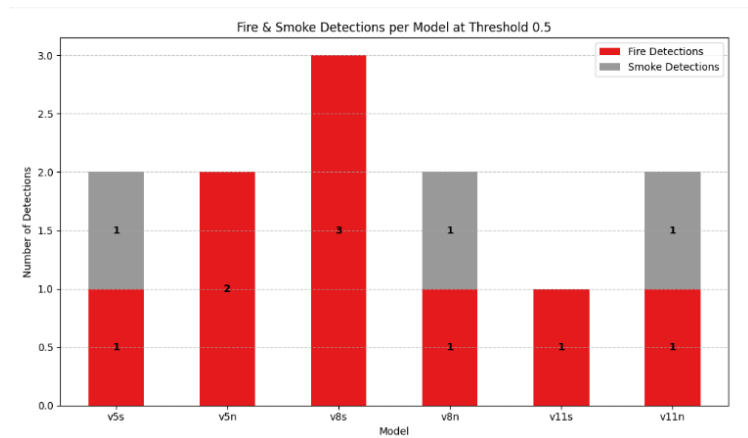


Fig 4.9: Fire & Smoke Detections per Model at Threshold 0.5

At Threshold 0.7

Every model at the 0.7 confidence threshold produced substantial decreases in detection outcomes as shown in Fig 4.10. The stringent threshold of 0.7 allows only confident predictions to pass through but simultaneously removes weak signals though it reduces the number of detected instances. YOLOv5s and YOLOv5n together with YOLOv8s and YOLOv8n along with YOLOv11s and YOLOv11n recognized at least one fire instance while YOLOv8n provided three successful detections. All models had minimal smoke detection rates because they concentrated on providing only highly certain predictions which caused their sensitivity to decrease.

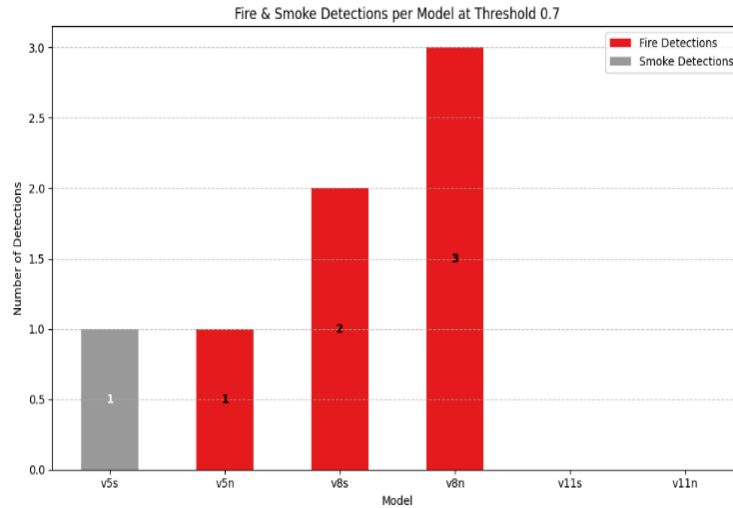


Fig 4.10: Fire & Smoke Detections per Model at Threshold 0.7

Overall YOLOv5s at 0.15 confidence threshold generated the best results in terms of balanced detection reliability of visual objects. The detection numbers from other models including YOLOv8n were higher at the 0.15 threshold yet YOLOv5s outperformed by creating precise fire and smoke bounding boxes which precisely placed objects without producing many false positive alerts.

This picture processed by YOLOv5s at the established threshold revealed both fire elements and smoke patterns despite changes in lighting levels and environmental backgrounds. The model delivered five accurate fire detections and one correct smoke detection on a non-training scenario as it generalized effectively to realistic settings.

This image produced reliable fire and smoke detection data by YOLOv5s which became the optimal choice for real-time fire detection systems because of its consistent and precise results as shown in fig 4.11.

```

Using device: cuda
Loaded Model Classes: {0: 'Cloud', 1: 'bonefire', 2: 'fire', 3: 'fog', 4: 'no fire-smoke', 5: 'smoke'}
Test image found: /content/45cf3484-a76f-49c8-99ac-80de467264c1_medium.webp
Performing Fire & Smoke Detection...

image 1/1 /content/45cf3484-a76f-49c8-99ac-80de467264c1_medium.webp: 384x640 5 fires, 1 smoke, 46.0ms
Speed: 2.7ms preprocess, 46.0ms inference, 5.4ms postprocess per image at shape (1, 3, 384, 640)
Results saved to runs/detect/predict7

Detection Results:
Class: smoke, Confidence: 0.76, Coordinates: (0, 0, 319, 141)
Class: fire, Confidence: 0.60, Coordinates: (232, 124, 263, 135)
Class: fire, Confidence: 0.55, Coordinates: (106, 125, 163, 143)
Class: fire, Confidence: 0.48, Coordinates: (193, 121, 209, 132)
Class: fire, Confidence: 0.18, Coordinates: (244, 124, 264, 131)
Class: fire, Confidence: 0.18, Coordinates: (235, 124, 263, 132)

```



Fig 4.11: YOLOv5s Detection Output at 0.15 Confidence Threshold

4.5 Real-Time Deployment on Raspberry Pi

The YOLOv5s model transitioned into TorchScript format for real-time detection deployment on Raspberry Pi 4B. The deployment system started by using the Pi Camera Module to obtain images while drones conducted their surveillance activities. The YOLOv5s model processed the frames that came from acquisition. Raspberry Pi activated the GPS module to automatically record the detection event coordinates after a detection event triggered its system. Raspberry Pi used its integrated Wi-Fi capability to send data consisting of the observed image with its class tag and GPS position to the FireGuard web platform. The implemented model operated at an average inference speed between 1.5 to 2.5 seconds per picture for real-time fire and smoke detection during deployment.

4.6 Web Portal (FireGuard System)

The deployment included the development of FireGuard which represents a functional web portal with real-time functionality. The portal allows users to visualize fire and smoke detection events in real-time and monitor GPS locations and enables access to historical detection logs and creates downloadable reports as shown in Fig 4 .12, 4.13. FireGuard was designed as a web portal featuring frontend and backend parts which supports multiple users for field operatives and control room staff to operate via any device.

4.6.1 Frontend Development

The FireGuard frontend operates with modern and extendable technological components which deliver swiftness together with modular functionality and instant response. The interface benefits from lightning-fast loading speeds because it was developed using Next.js which is a React-based framework that offers both server-side rendering and API route functionalities. React.js components were created using React.js but the TypeScript integration brought additional reliability through static type checking functionality. Tailwind Cascading Style Sheets (CSS) served as the styling solution because it delivered utilitarian framework capabilities which promoted streamlined design together with design consistency. The development process was accelerated through the use of shadcn/ui which provides ready-made components for alert cards and forms alongside tables. Lucide React implemented visual elements as a modern and lightweight system through its visual language design.

Several vital features appear in the frontend section. The system combines Firebase Authentication with its user roles function which grants admin permission and analyst permission and viewer permission for secure login access control. Users can access FireGuard dashboard for real-time active detection summary together with system performance indicators and historical statistical data. The system shows incoming alerts in real time through a live detection view which includes the class information along with timestamps and model confidence percentiles. The FireGuard system displays fire-smoke events on a map interface through Google Maps API that uses severity-based color markers to indicate their positions on the map. Users can view past

detection records through the detection history page that enables date-based and class-based and location-based filters. Users can utilize the reports section to generate detection data both in Portable Document Format (PDF) and Comma Separated Values (CSV) file formats for documentation purposes alongside regulatory needs. All users can access the responsive interface through desktop and tablet and mobile devices which fits field operations.

4.6.2 Backend Development

The backend implementation of FireGuard utilized Flask as its Python web framework because it provides REST API development and enables easy connection between Firebase and Raspberry Pi. The backend system operates as the foundation which controls data input management along with user login features and updates real-time database records and establishes network communication between embedded hardware and web portal elements.

The application endpoints built through Flask processed HyperText Transfer Protocol (HTTP) requests from the Raspberry Pi device after detecting fire or smoke inside manufacturing facilities. The API requests deliver information which includes the analyzed image and its predicted category alongside position coordinates and accuracy value and timestamp. The system stores data points in the Firebase Realtime Database to make them immediately available for the frontend interface. User authentication takes place through Firebase Authentication to validate identities across multiple features through Role-Based Access Control (RBAC) permissions systems.

The backend system processes timestamps while formatting GPS data and storing upload image paths and manages the database structure. The system uses Firebase triggers for live dashboard synchronization which shows new detections immediately on the frontend components. Through the API system users gain access to downloadable PDF and CSV reports and logs. An optional alert system exists to manage critical events through automated email and Short Message Service (SMS) notifications that use configurable services. FireGuard operates with reliable real-time functionality because its robust backend system remains easy to maintain along with scalability and security features.

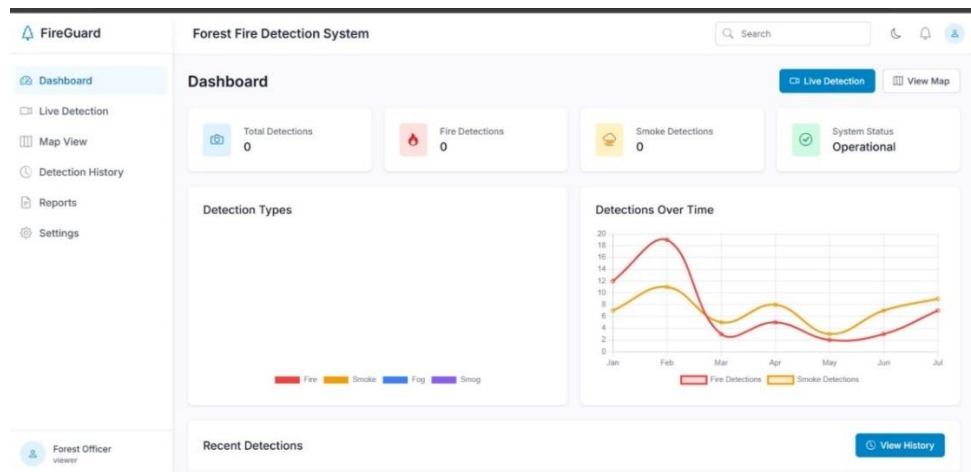


Fig 4.12: FireGuard web portal showing the dashboard

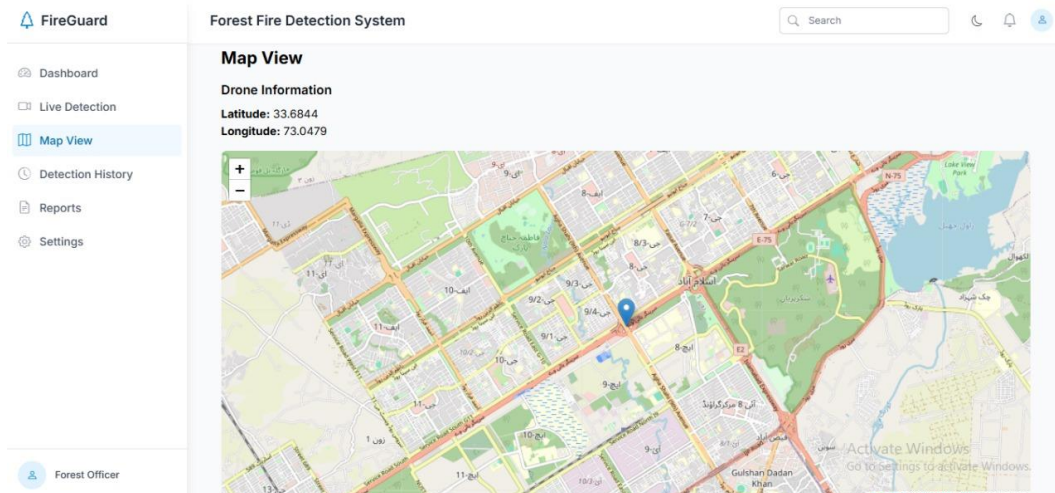


Fig 4.13: Web portal (Map view)

CHAPTER 5

Conclusion

Forest ecosystems maintain planetary life while supporting biodiversity through their climatic regulation and their supply of endless natural resources. Natural disasters ranked forest fires in the top destructive category because these events wreak damage on ecosystems and properties as well as endanger human existence. The growing threats from human-caused activities, together with climate change factors, have dramatically increased the number of wildfires to dangerous levels, thus making early detection and immediate response systems crucial.

The presented research seeks to create a cost-efficient fire detection system that uses drone technology with image processing methods for early fire alarm capabilities. The main objective revolved around spotting emerging fire, together with smoke, fog, or clouds, in addition to precisely locating positions via GPS for prompt warning systems and swift emergency teams. A manually controlled drone carried multiple technologies to detect early signs of fire by employing a Raspberry Pi camera module and Raspberry Pi 4 together with YOLO and GPS modules. The processed data that emerged through Python scripts got presented through a web-based platform.

The YOLO object detection algorithm successfully recognized fire, together with smoke and clouds, as well as fog in static images during implementation. The combination of high processing speed and real-time detection capabilities, together with precision, made YOLO an ideal choice for low-powered Raspberry Pi devices. GPS allowed the system to send accurate detection location data which strengthened both emergency rescue and localization speed after an event.

The main advantage of this research project emerges from its direct application in real-world scenarios. We intentionally made it both easy to implement and affordable so that it could work in resource-limited regions, including remote forests in Pakistan. Testing of the designed prototype yielded successful results as it detected fire

indications together with smoke as well as fog and clouds in real-time scenarios. Future system enhancements and integrated advanced capabilities can be added through the modular design while maintaining the existing structure without performing a complete revamp.

Through its user interface, emergency staff and forest rangers obtained real-time GPS tracking capabilities combined with image-based surveillance for remote forest inspection. This technology provides fast detection and minimizes response time, which prevents widespread destruction.

This research proves that reasonably priced hardware paired with open-source tools enables the development of an efficient ‘Early Fire Detection in Forest using Image Processing’. The device demonstrates its value to environmental conservation through its accessible smart design, which proves that embedded systems technology, together with machine learning, can tackle real-world issues.

CHAPTER 6

Future Work

Subsequent improvements can be made to the existing platform to optimize its operational robustness and efficiency as well as scalability capabilities. The system requires ongoing development because forest environments show high levels of unpredictability and diversity.

Under low-light settings the RGB camera provides sufficient data for detecting flames and smoke observations, but its performance reduces significantly. The system can identify heat signatures thanks to thermal cameras when flames are invisible to the naked eye. By utilizing multispectral sensors such as those that measure data beyond the human viewable spectrum the system can produce better fire activity detection results.

The YOLO deep learning model requires extensive computing resources due to its processing demands during embedded system real-time operations. Subsequent research should investigate compression methods including quantification and waveform removal because they lower model dimensions without diminishing predictive performance. Edge processing speed can be enhanced through the implementation of platforms including NVIDIA Jetson Nano and Google Coral.

Manual operation of the drone currently restricts both its operational area and growth capabilities. The drone system requires development of autonomous navigation through GPS waypoint tracking combined with obstacle detection algorithms to achieve automated forest patrols. The system will enable surveillance over larger areas because human operator presence will not be required for continuous control.

Fog together with sunlight reflections and dust particles generate frequent false alarm triggers in fire detection systems. The forthcoming system releases might include context-optimized algorithms which combine various environmental factors such as temperature and wind direction together with humidity to boost detection integrity.

The current system went through validity testing that took place in artificial environments. Tests under real-world forest environments with different terrain types and environmental variations need to be performed to advance the system for widespread adoption. The planned tests will allow researchers to establish exact thresholds for detection and perform system adjustments according to different environmental zones.

An automated emergency alert system that streams real-time information to stakeholders can be enabled by integrating the detection system with existing public or regional emergency response frameworks. The system can trigger time-sensitive alerts that send messages through SMS, emails or mobile apps to stakeholders to enhance their response speeds and resource management.

A mobile application designed specifically for this purpose would grant users real-time monitoring, automated alert systems, live streaming feeds as well as geographic tracking capabilities. The mobile app would help forest rangers and disaster response teams monitor incidents while they perform their field activities.

The next generation should include built-in solar panels together with energy-harvesting technology to power both the drone and Raspberry Pi system which would enable prolonged operations without recurring battery substitutions in remote locations.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI:	Artificial Intelligence
ANN:	Artificial Neural Network
API:	Application Programming Interface
CNN:	Convolutional Neural Network
COCO:	Common Objects in Context (Dataset)
CPU:	Central Processing Unit
CSI:	Camera Serial Interface
CSS:	Cascading Style Sheets
CSV:	Comma-Separated Values
DB:	Database
DJI:	Da-Jiang Innovations (Drone Manufacturer)
DL:	Deep Learning
FCM:	Firebase Cloud Messaging
FLAME:	Fire Luminosity and Appearance Metadata for Evaluation
FPS:	Frames Per Second
GPU:	Graphics Processing Unit
HTTP:	HyperText Transfer Protocol
IR:	Infrared
IoT:	Internet of Things
IOU:	Intersection Over Union
JS:	JavaScript

Abbreviation	Full Form
LiDAR:	Light Detection and Ranging
LoRaWAN:	Long Range Wide Area Network
LR:	Learning Rate
mAP:	mean Average Precision
MFA:	Multi-Factor Authentication
ML:	Machine Learning
MQTT:	Message Queuing Telemetry Transport
MST:	Minimum Spanning Tree
NMEA:	National Marine Electronics Association
ONNX:	Open Neural Network Exchange
OpenCV:	Open-Source Computer Vision Library
OS:	Operating System
PDF:	Portable Document Format
PI:	Raspberry Pi
PSO:	Particle Swarm Optimization
PTZ:	Pan-Tilt-Zoom (Camera)
PyTorch:	Python Torch (Deep learning framework)
RAM:	Random Access Memory
RBAC:	Role-Based Access Control
RCNN:	Region-based Convolutional Neural Network
RF:	Random Forest
R-FCN:	Region-based Fully Convolutional Network

Abbreviation	Full Form
RGB:	Red, Green, Blue
ROS:	Robot Operating System
SDK:	Software Development Kit
SGD:	Stochastic Gradient Descent
SSD:	Single Shot MultiBox Detector
SVM:	Support Vector Machine
UART:	Universal Asynchronous Receiver-Transmitter
UAV:	Unmanned Aerial Vehicle
UNet:	U-shaped Neural Network
VIIRS:	Visible Infrared Imaging Radiometer Suite
Wi-Fi:	Wireless Fidelity
WSDY:	Wildfire Smoke Detection from YouTube
WSN:	Wireless Sensor Network
WSN:	Wireless Sensor Network
WWF:	World Wide Fund for Nature
XGBoost:	Extreme Gradient Boosting

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